REED'S MARINE ENGINEERING SERIES

10

INSTRUMENTATION AND CONTROL SYSTEMS

DATE DUE

REED'S INSTRUMENTATION AND CONTROL SYSTEMS

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PREFACE

This book aims to bridge the gap between the mathematical treatment often used by the specialist control engineer and the (necessarily) narrow descriptive literature of a particular manufacturer.

It is written primarily for those with a good general engineering background who have had little experience in instrumentation and control.

The work favours marine engineering but students and engineers in other industries should find it a useful book as the subject has a common basis. Text and examples cover the requirements of all Department of Transport (DTp) – and all Business and Technician Education Council (BTEC) and SCOTVEC – syllabuses and examinations for marine engineer officers and cadets.

Full use has been made of simplified diagrams and the work is presented from basic principles, using analogues where appropriate.

The introduction is followed by five chapters on variable measurement in instrumentation. Chapters 6–8 on telemetering, electronic devices and final controlling elements link instrumentation to control. Chapters 9–14 cover theory, practice and components of process and kinetic control systems. Chapter 15 and 16 are intended to develop a broader knowledge of the subject and, by necessity, have a more analytic and mathematical approach.

A selection of test examples are included at the end of each chapter and specimen examination questions are added at the end of the book.

If the reader wishes to obtain a full detailed description of a particular component reference should be made to manufacturers' instruction manuals.

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INTRODUCTION

HISTORICAL

Instrumentation has always been an integral part of technology. Development from simple level indicators, Bourdon tubes, moving iron and moving coil meters, etc. has been rapid. Progress in electrics and electronics has led to centralised recording and display stations with associated data processing, computing and control systems. Application to control with the requirement of accurate measuring (sensors), variable converting (transducers) and remote signal transmission (telemetering) has involved a close relation between measuring, processing and control systems. The advantage of electrical signal transmission is apparent in the development of instrumentation. Modular designs and interchangeable plug-in sub-assemblies have improved servicing of electronic units. Digital operation is increasingly being favoured over analogue operation.

The development of control elements is inherent in the history of man as life form itself utilises classic control principles. The Watt governor (1788) was one of the first practical applications. Instability was recognised last century in hunting of steam engine speed, and ship steering gears, and much analysis followed. The ship steering gear remains as one of the first control systems and its development alongside modern bridge equipment continues. By 1939 fairly complex systems, mainly pneumatic and hydraulic, were in use and development in electronics, related to the feedback amplifier, has accelerated progress and now leads to computer control. Systems are generally classified by their field of operation. Process control such as flow, level and pressure; kinetic control such as displacement, velocity and acceleration; etc.

UTILISATION

Automation is essentially sequence-controlled mechanisation and is only an aspect of control. The word tends to varied usage and as such is not considered further in this text. American terminology tends to use the word *cybernetics* to describe the entire field of communication and control theory. A full study programme of the subject, which is generally followed in this text, is shown in the first sketch.

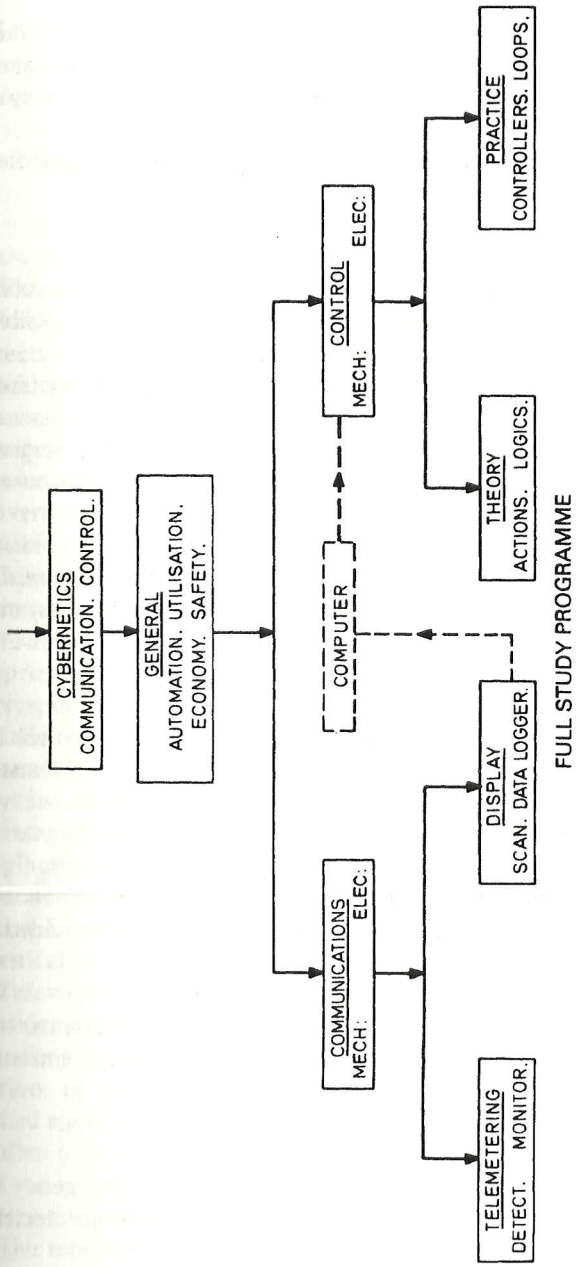
The degree of utilisation in marine practice varies a great deal. Individual control loops, from simple to fairly complex, have been in use for many years. Centralised data handling has been a more recent innovation. Bridge control and unmanned machinery spaces have developed quite rapidly with improved reliability of control and alarm loops. The next stage of development is to link the centralised data handling system to an integrated central control system. This requires that a computer is involved in the interface between measurement and control. Computer control has developed most recently, from small programmed functions to quite sophisticated direct digitally controlled processes. A modern computer can be so programmed not only to control machinery under all conditions but also to have start up, emergency and shut down procedures. Extension to navigational, cargo handling, etc. easily follows.

ECONOMY

Instrumentation and control results in more efficient operation and reduced manpower in every case. There is an increased first cost due to specialised equipment provision which leads to increased insurance requirements and some increase in certain running costs, e.g. staff training, skilled maintenance, etc. Overall running costs are reduced because of large cost savings in fuel and general maintenance, due to efficient operation and close supervision, as well as staff reductions. The annual saving taking all factors into account is well proven for controlled plant and the factor increases with increasing size of plant and machinery.

SAFETY

In most cases safety is improved by monitoring and control. Reliability of both measuring-control alarm equipment is essential



as indeed should be the characteristics of the plant itself. These aspects should be analysed closely at the design stage. Unmanned machinery spaces (UMS) are now classified by surveying societies.

Essential requirements for unattended machinery spaces could be summarised thus:

1. Bridge control of propulsion machinery.

The bridge watchkeeper must be able to take emergency control action. Control and instrumentation must be as simple as possible.

2. Centralised control and instruments are required in machinery space.

Engineers may be called to the machinery space in emergency and controls must be easily reached and fully comprehensive.

3. Automatic fire detection system.

Alarm and detection system must operate very rapidly. Numerous well sited and quick response detectors (sensors) must be fitted.

4. Fire extinguishing system.

In addition to conventional hand extinguishers a control fire station remote from the machinery space is essential. The station must give control of emergency pumps, generators, valves, ventilators, extinguishing media, etc.

5. Alarm system.

A comprehensive machinery alarm system must be provided for control and accommodation areas.

6. Automatic bilge high level fluid alarms and pumping units.

Sensing devices in bilges with alarms and hand or automatic pump cut in devices must be provided.

7. Automatic start emergency generator.

Such a generator is best connected to separate emergency bus bars. The primary function is to give protection from electrical blackout conditions.

8. Local hand control of essential machinery.

9. Adequate settling tank storage capacity.

10. Regular testing and maintenance of intrumentation.

TERMINOLOGY

The *detecting element* responds directly to the value of the variable. A *measuring element* responds to the signal from the detecting element and gives a signal representing the variable value. For example pressure (variable), Bourdon tube (detecting element) and linkage pointer, scale, *i.e.* pressure gauge (*measuring element*). *The measuring unit* comprises detecting element and measuring element. Such a unit is used as a *monitoring element* (to convert, when necessary, the actual variable value into a converted variable value) of a process control system

Sensor (American) is a term used for the detecting element. Is, by its very nature, essentially a *transducer*.

Transducer (American) is a device to convert a signal (representing a physical quantity) of one form into a corresponding signal of another form, retaining the amplitude variations of energy being converted. For example a microphone is a sound transducer (acoustic to electrical) and a loud speaker an electrical transducer (electrical to acoustic). A transducer may be an integral part of the measuring unit, for example pressure to displacement in a Bourdon pressure gauge. It may also be a separate unit converter especially suitable to change the signal to a better form for remote transmissions, e.g. displacement to electrical in a differential transformer.

Telemetering may be defined as signal transmission over a considerable distance. In measurement this involves information transfer from detecting element to a central recording-display station. In control this involves control operating devices and related signal transfers. In telemetering systems the measuring unit is often called the *transmitter*, usually incorporating a transducer, and the recording unit some distance away is then referred to as the *receiver* which may have an associated transducer if required.

The terminology involved further to the above and especially

related to control systems is now fairly extensive. Such terminology is covered in some detail at the start of Chapter 9.

Instrumentation in this book is generally confined to dynamic systems related to recording and control. Obviously the range is much wider if extended to include static-laboratory type instrument devices.

COMPARISON OF SYSTEMS

Systems, telemetering or control, may be either pneumatic, hydraulic, or electronic-electric, or a combination.

Hydraulic systems are generally more restricted in application. Basically the technique is as for pneumatics but fluid cannot be allowed to escape and a recovery-storage system is required. General use is in the higher pressure range.

A combination of electronic measure-record instrumentation and pneumatic final power control element is very effective. Controllers may be either pneumatic or electronic. The former have generally been used because of proven reliability and ease of application to final power transmission. Electronic controllers are increasingly being used and the all electronic-electric system has many obvious advantages.

The advantage of a pneumatic system are:

1. Less expensive initially, this is in spite of tubing and air supply costs.
2. No heat generation and safe in explosive atmosphere.
3. Less susceptible to power supply variation, but do have appreciable time lags.
4. Direct application, without transducers, to large final power actuators.

The advantages of an electronic system are:

1. Small and adaptable with cheap flexible transmission lines.
2. No moving parts, can however generate heat.
3. Stable, generally accurate and very short time lags.
4. Low power consumption, direct application to computers, but often need final element transducers.

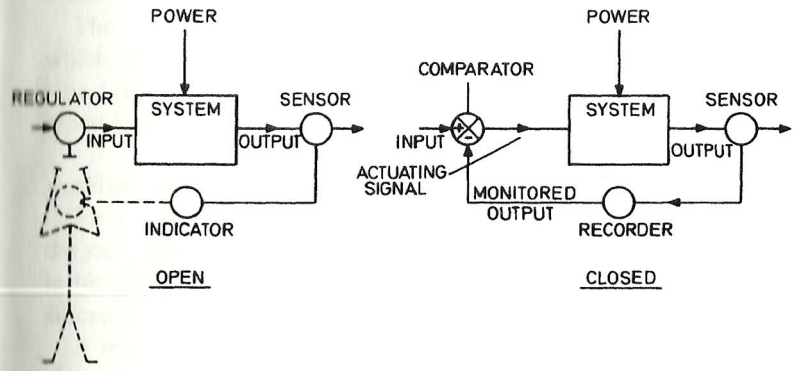
In every case equipment must be robust, reliable, interchangeable,

simple and resistant to environment. A long commissioning time should be applied and regular skilled maintenance is required. Signal dc transmission is usually preferred although ac signals are essential for certain variables and easy amplification of ac is an inherent advantage.

CONTROL LOOPS

An open loop system has no feedback and controller action is not related to final result. Consider a domestic central heating system as an example, in which fuel supply is varied manually or automatically by external ambient temperature. Room temperature will be maintained at a reasonable value related to outside conditions. However room temperature does not control fuel supply so that this is open loop. The word loop is really a misnomer.

Now to the open loop shown, add human operator, so closing the loop (dotted lines on sketch). This is a manually controlled closed loop system.



CONTROL LOOPS

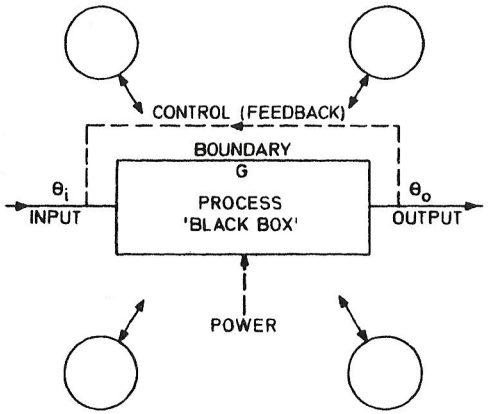
The automatically controlled closed loop system is actuated by a signal dependent upon deviation (error) between input (set) and output values. Deviation only exists when monitored output (negative feedback) differs from input and this signal controls power supplied to output. For a closed loop system, as sketched, output power is only controlled by, and not supplied by, the actuating signal. Closed loops have a self regulating property.

SYSTEM

An assembly of linked components within a boundary. The motor car is a good example; mechanical, electrical, control and suspension sub-systems within a body-chassis boundary. A system may have one input and a related output dependent on the effect of that system (**transfer function** *G*).

$$\theta_0 = G\theta_i$$

The boundary, represented as a"black box", may include a complex system which need not be analysed if G is provided. More complex systems have interconnecting links to related systems. A system must have input, process, output, and in most systems a source of power and a means of control.



SYSTEM

ANALOGUE

Many different physical phenomena behave in a similar way, i.e. are analogues of each other. Two examples are air escape from a storage vessel and electrical charge loss from a capacitor.

Rate of change of pressure α pressure
Rate of change of charge α voltage

i.e. Rate of change of variable $\dot{x} = -kx$

$$\frac{dx}{dt} = -kx$$

A solution, where C is the x value when time is zero, is:

$$kt = l_n\frac{C}{x}$$

Voltage (or current) can readily be made analogous to physical phenomena. The traditional electric clock is analogue *i.e.* continuous representation.

DIGITAL

A digital device manipulates "bits", *i.e.* discrete items of information - illustrated by the digital clock representation. States are on/off, equal/unequal, etc. and the binary digit system is utilised.

COMPUTERS

The electronic analogue computer is essentially a simulator on which electrical analogues of various systems can be analysed and illustrated. The digital computer is a machine for routine, repetitive arithmetic. Hybrid types are a combination.

MICROPROCESSORS AND MICROCOMPUTERS

First generation computers (approx. room size) were often, for the same capacity, replaced by minicomputers (say desk size) and in turn by microcomputers (hand size and smaller) following the silicon chip and integrated circuit (IC) development.

A microprocessor (μP) – component product – may be a single chip unit or a collection of unassembled processor-related components such as central processor unit (CPU), timers, memories, interfaces, etc.

A microcomputer (μCP) – board product – assembly of μP components mounted on a printed circuit board sufficient to make up a working computer.

Semi-conductor and metal oxide silicon (MOS) – random access memory (RAM) and read only memory (ROM) are utilised for both CPU and logic system μP.

Whilst the μP, consisting of arithmetic logic units plus sets of registers and control circuits, cannot be used by itself to create a system it can, with support chips, form a dedicated controller or μCP.

CHAPTER 1

TEMPERATURE MEASUREMENT

This chapter is concerned with the practical aspects of thermometry. No consideration is applied to absolute standards or to the consideration of special techniques related to extreme temperatures , etc.

MECHANICAL THERMOMETRY

LIQUID IN GLASS THERMOMETER

Mercury can be used from –38°C (its freezing point) to about 600°C. For the higher temperatures an inert gas at high pressure is introduced as the boiling point of mercury is about 360°C at atmospheric pressure; special glass is also required.

Alcohol is used in the range –80°C to 70°C (or toluene) and pentane can be used to –196°C.

Total immersion types are most accurate, especially when the fluid is coloured and magnification is used. In many cases only temperature *differential* is required so that relatively low accuracy partial immersion types are often satisfactory.

FILLED-SYSTEM THERMOMETER

Consist of liquid, vapour or gas filled types. All utilise a bulb, connecting capillary and usually a Bourdon tube mechanism, responding to pressure change from volume variation (liquid), for pointer or pen operation. Some systems incorporate a compensating capillary and bourdon tube to allow for changes of ambient temperature. Alternatively a bi-metallic link for compensation can be incorporated into the mechanism.

A common type of liquid filled system utilises mercury in steel which can be pressurised for high temperature duty to 600°C. Such devices are most useful for remote sensing and telemetering back to a central instrumentation panel. Capillary bore is about 0·02 mm, the scale is generally linear but calibration must allow for heat variation. Power is sufficient for pointer, pen or transducer operation.

Vapour pressure thermometers commonly use freon, alcohol or ether which partly fills the system as liquid, and the remainder is vapour filled. Measurement of vapour pressure gives an indication of liquid surface temperature and is usually used in the range –50°C to 260°C, with the upper limit fixed by the critical temperature of the liquid which must have a low boiling point (high vapour pressure). The scale is non-linear, ambient variations can be neglected but there can be appreciable time lags and the device is not well suited to remote indication.

Gas filled thermometers usually employ nitrogen or helium under high pressure, and pressure is proportional to absolute temperature at constant volume. The usual temperature range is –50°C to 430°C and the scale is linear. Compensation for ambient temperature variation is difficult. When used as a sensor linked to a pneumatic transducer it is a very effective device.

BI-METALLIC THERMOMETER

The principle of operation of bi-metallic devices is that of differential expansion of two different materials rigidly joined together, one on the other, as a strip of bi-metallic material. Figure 1.1 illustrates a typical design usually employed between –40°C and 320°C. Invar (36% Ni, 64% Fe) has a low coefficient of expansion and when welded to a Ni-Mo alloy gives a good bi-metallic strip. The helix coils or uncoils with temperature variation and as one end is fixed the movement rotates shaft and pointer. The range of the instrument is fixed by the materials used.

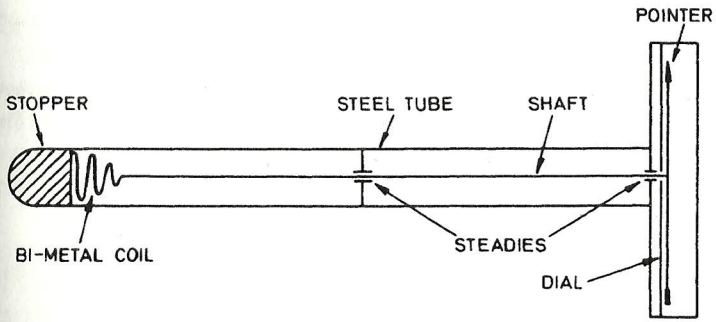


Fig. 1.1 MECHANICAL TYPE THERMOMETER (BI-METALLIC TYPE)

ELECTRICAL THERMOMETRY

RESISTANCE THERMOMETER

The electrical resistance of a metal varies with temperature and this relationship is usually expressed as $\rho_\theta = \rho_0 (1 + \alpha\theta)$

where ρ_θ is the specific resistance at temperature θ°C

" ρ_0 is the specific resistance at temperature 0°C

" α is a constant which depends upon the metal and is called the temperature coefficient of resistance.

Figure 1.2 shows diagrammatically a resistance type of temperature measuring unit using the well known Wheatstone bridge principle. r_1 r_2 is a variable resistance used for balance purposes; at balance we have:

$$\frac{R_1 + r_1}{R_2 + r_2} = \frac{R_3 + r}{R_4 + r}$$

r is the resistance of each of the wires and since each wire will be subjected to the same temperature variation along its length their resistances will always be equal.

When the temperature detecting element is subjected to a temperature alteration its resistance alters and the bridge balance is upset. By using the variable resistor r_1 r_2 balance can be restored (i.e galvanometer reading returned to zero) and whilst this is being done another pointer can be moved simultaneously and automatically to give the temperature – this is known as the null

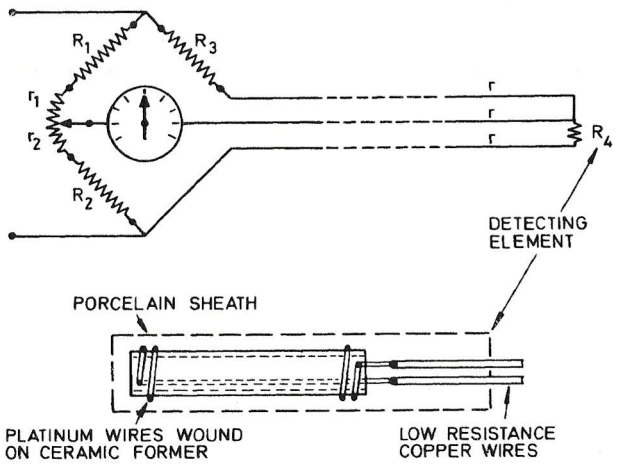


Fig. 1.2 RESISTANCE THERMOMETER

balance method Alternatively, the galvanometer can give the temperature reading directly, in this case no variable resistance r_1 r_2 would be required.

For the measurement of ambient temperature conditions the resistances, apart from the temperature measuring resistance, would have to be made of a metal whose resistance does not vary with temperature. A metal which nearly fulfills this requirement is constantan.

The term pyrometer is used for temperature measuring instruments operating above 500°C, the term thermometer being used for operation below 500°C.

Resistance thermometers can be exceedingly accurate. Platinum is the most suitable sensing wire element but copper and nickel wire are used in the range −100°C to 200°C and tungsten, molybdenum and tantalum are used to 1200°C, in protective atmospheres. The platinum element usually has a resistance of 100 ohms at 0°C in which case resistance of wires is limited to about 3 ohms. Use up to 600°C with twin wires is often acceptable with the three wire method used for higher accuracy; measurement is by Wheatstone, Kelvin or Mueller bridges or potentiometric methods.

THERMISTOR (THERMALLY SENSITIVE RESISTOR)

These devices are among a second class of resistance thermometer utilising elements made of semi-conducting material all of which have a characteristic of a resistance decrease with temperature increase. Included in this category are carbon resistors and doped germanium units (see Fig. 1.3).

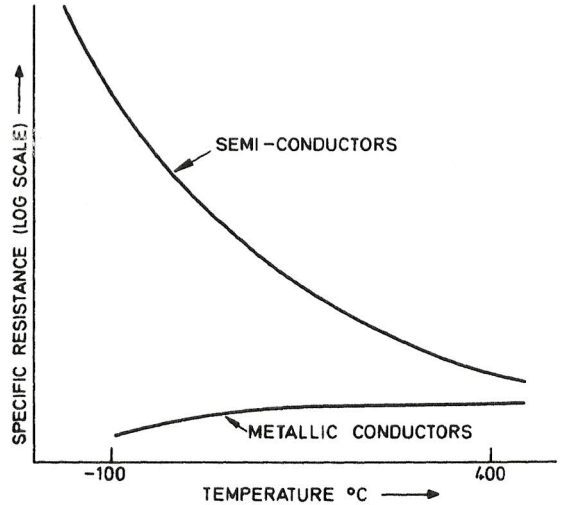


Fig. 1.3 RESISTANCE-TEMPERATURE CHARACTERISTICS

Thermistors are made of semi-conducting materials, they are manufactured by sintering (i.e. heating under pressure) powder mixtures of metallic oxides such as manganese, nickel, cobalt, copper, iron or uranium. The size and configuration can be controlled so that rods, beads, discs and washer shapes can be produced as desired. Figure 1.4 shows an ellipsoid of thermistor material, wires about 0·25 mm apart are firmly embedded in the material making good electrical contact. The whole assembly may be coated with glass to give strength and protection.

A washer shape of thermistor is particularly useful, it could be fitted over bolts or studs. Several washers could be used together with the terminals connected in series or parallel arrangements. Beads or rods of thermistor material are suitable for use as probes.

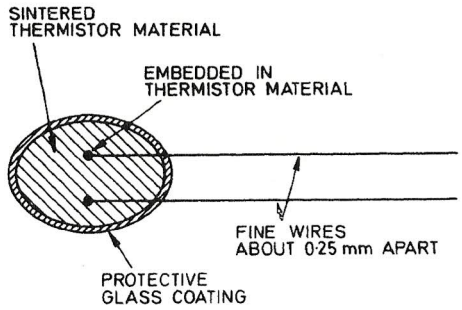


Fig. 1.4 THERMISTOR BEAD

Advantages of thermistors:

(1) Relatively small and compact, the bead arrangement shown could have a diameter up to 2·5 mm with a resistance up to about 100 megohms.

(2) Low specific heat, hence the thermistor does not take very much heat away.

(3) Physically strong and rugged.

(4) Relatively high temperature coefficient of resistance, it could be as high as ten times that of some metals.

(5) They can be used for extremely low temperature measurement with great accuracy.

The mathematical relationship for thermistors is given by:

$$\rho_\theta = \rho_0 e^{\beta\ (1/\theta\ -\ 1/\theta_0)}$$

where:

ρ_θ is the specific resistance at temperature θ

ρ_0 is the specific resistance at temperature θ_0

β is a constant which depends upon the material used in the construction ($\beta \simeq 4000$). The characteristic is shown in Fig 1.3.

The large negative temperature coefficient of resistance of thermistors may be explained by considering the number of electrons available for carrying current.

Few electrons are available at low temperatures but as the temperature increases the kinetic energy of the electrons increases

and this enables them to move from inner tightly bound orbitals to the outer conduction bands of the atom. With more free electrons available to carry current the resistance to current flow reduces.

In metals, where there are many free electrons, increase in temperature leads to "traffic jams" of electrons hence the resistance to current flow increases with increase of temperature.

The very high sensitivity, small size, high energy and rapid response makes thermistors very useful detecting devices particularly for use in computers and scanner installations. Range is –100°C to 300°C but special compositions can extend this much further, as high as 1600°C. The small thermal mass can lead to self heating and coupled with high sensitivity and exponential characteristic means instability must be carefully watched. The thermistor merely replaces the resistance element on one limb of a bridge circuit in the measuring unit.

THERMOCOUPLE

Whenever the junctions formed of two dissimilar homogeneous materials are exposed to a temperature difference, an emf will be generated which is dependent on that temperature difference, also on the temperature level and the materials involved. This thermoelectric emf is called Seebeck effect (Seebeck discovered in 1821) and is an algebraic sum of two other effects discovered by Peltier and Thomson. The two materials, usually metals, form the thermocouple.

Figure 1.5 shows a thermocouple consisting of two wires, one iron, one constantan (*i.e.* a copper nickel alloy), with a millivoltmeter coupled to the copper wire. If the junctions A and B are maintained at the same temperature no current will flow around the circuit since the emfs in the circuit will be equal and opposite. If however A is heated to a higher temperature than B then current will flow since the emf at one junction will be greater than the opposing emf at the other junction.

A third wire can be introduced as shown in Fig. 1.5, where AB and AC form the couple wires. Providing the junctions B and C are maintained at the same temperature, the introduction of the third wire BC will not affect the emf generated. Hence A will be the hot junction and B with C will form the cold junction. Couple wires AB and AC shown as iron and constantan respectively can

be made of various metals and alloys, choice depends upon temperature of operation, the wire BC would generally be long compared to the couple wires and could be made of copper. Figure 1.5 also shows the device in detail.

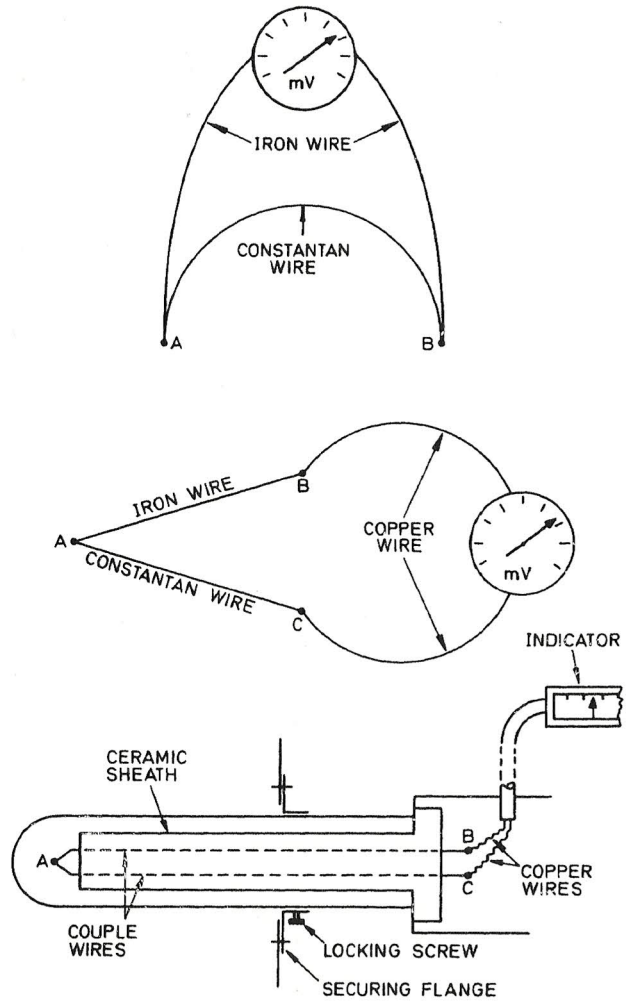


Fig. 1.5 THERMOCOUPLE

In a practical thermocouple system the cold junction B and C may be at a relatively high temperature due to the environment. This would mean that the temperature difference between the hot and cold junctions would be small and mV similarly. The indicator itself could then become the cold junction if the wires from terminals B and C to the indicators are of the same material, or material with similar characteristics, to the couple wires. The wires are then called compensating wires and the cold junction temperature would be reasonably constant if the indicator is within an air conditioned control room or immersed in a block of metal of large thermal capacity. Alternatively cold junction compensation signal by separate means or bi-metallic instrument components is arranged.

A copper (+) constantan (–) couple is used up to about 350°C, constantan being a 40% Ni 60% Cu alloy. Up to 850°C an iron-constantan couple is used with a chromel (90% Ni 10% Cr) and alumel (94% Ni 2% Al) couple up to 1200°C. Average emf is 0·05mV/°C which compares with about 18mV/°C for a thermistor. Platinum-platinum plus 10% rhodium couples have been used to 1400°C.

The emf generated is usually given by an expression of the form:

$$e = A + B\theta + C\theta^2 + D\theta^3 + \cdots$$

where θ is the temperature and A, B, C, D, are constants of diminishing order. Fig. 1.6 shows an electronic thermocouple with operational amplifier (A) giving 0.1 V/°C. Y is for calibration at cold junction temperature and X for other temperature calibration.

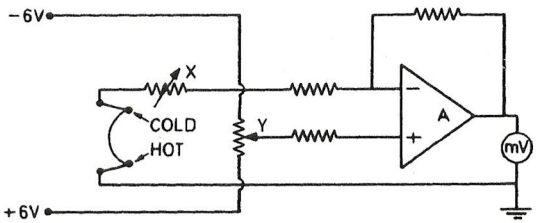


Fig. 1.6 ELECTRONIC THERMOCOUPLE

RADIATION PYROMETRY

When temperatures are above the practical range of thermocouples, or the "target" is not accessible, or an average temperature of a large surface is required then radiation pyrometers are used. Theory is generally based on black body radiation and the work of Stefan, Boltzmann and Planck with amended factors of emissivity to allow for variation from the ideal black body radiator. Types of radiation pyrometer are optical, radiation and two colour. The former will be considered.

OPTICAL PYROMETER

Referring to Fig. 1.7: S is the source and rays enter lamp box L after passing through the lens, aperture and absorption filter. The lamp is electric and current and voltage are measured at G. Rays leaving L pass through a red filter, lens and aperture to eye E.

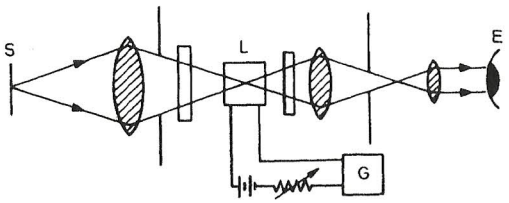


Fig. 1.7 OPTICAL PYROMETER

The device is often known as a disappearing filament unit. Both source and reference, the latter being the filament of a small vacuum lamp, are observed through the microscope. The power to the lamp is adjusted until the reference source just disappears into the main source. Power is calibrated to give a temperature reading directly. Correction factors apply for the filters used and the device is a selective radiation pyrometer as only a narrow band of radiation wavelength is utilised. Radiation pyrometry is particularly useful for furnace, molten metal, process control, etc. evaluation of temperature.

PHOTO-ELECTRIC PYROMETER

There are three types of photo-electric cells; the photo-emissive, the photo-conductive and the photo-voltaic; the latter is used here.

Incident light falls on p-type silicon layered on to n-type silicon backed with metallic strip. The emf generated is measured, after calibration, by a galvanometer or self balancing potentiometer connected across the p-type and the backing.

Such pyrometers are best suited to measuring small radiation sources and are stable and accurate with a very quick response time which makes them very suitable for distance remote control systems.

Note:

Many of the measuring devices for temperature considered previously, particularly electrical types, could also be classified as telemetering or transmitting units as the signal is readily conveyed over a considerable distance to a remote measuring, recording or display station. This applies in many cases to other such devices in the following chapters. The detecting element (sensor) is inherently a transducer in operation for many instrument units.

TEST EXAMPLES 1

1. Describe, with the aid of simple sketches, three types of temperature measuring device. State how they are graduated and where they are used.

2. Sketch and describe an electrical instrument used for reading temperature at a remote distance. State the usual temperature range and the materials used in construction.

3. Give a reasoned explanation of how Wheatstone bridge networks are employed in circuits of electrical resistance thermometers, explosive gas sampling devices or any similar application. Sketch the circuit for such a device indicating the function of the Wheatstone bridge.

4. Sketch and describe a temperature measuring system employing a thermocouple.

CHAPTER 2

PRESSURE MEASUREMENT

WATER MANOMETER

This instrument is used for measuring pressures of a low order such as fan pressures, etc. Fig. 2.1 illustrates a U tube water manometer, one limb of which is connected to the system whose pressure is to be measured, the other limb is open to the atmosphere. The pressure reading is the difference of the water levels read from the scale.

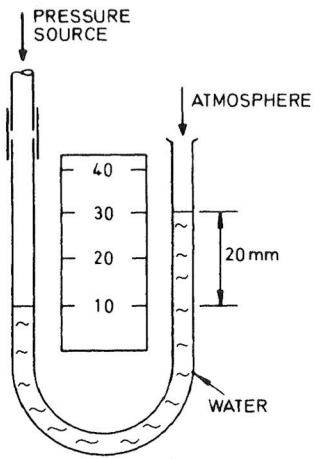


Fig. 2.1 WATER MANOMETER (U TUBE)

Note: 1 m^3 of fresh water has a mass of 1 Mg and weighs 10^3 x 9·81 N.

Hence:

1 m of fresh water exerts a pressure of 10^3 x 9·81 N/m^2, 1 mm of fresh water exerts a pressure of 9·81 N/m^2, the reading indicated in Fig. 2.1 is therefore equivalent to 20 x 9·81 = 186·2 N/m^2, i.e. 0·1862 kN/m^2, above atmospheric pressure (1 bar = 10^5 Pa = 10^5 N/m^2 ≃ 1 atm). A wide cistern manometer (well type) is used for lower pressure differentials (Fig. 2.2 as mercury manometer). A variation is the inclined tube manometer – the small bore limb is set at a small angle to the horizontal and the longer scale parallel to it gives even smaller differential pressure readings.

MERCURY MANOMETER

A mercury manometer of the well type is shown in Fig. 2.2.

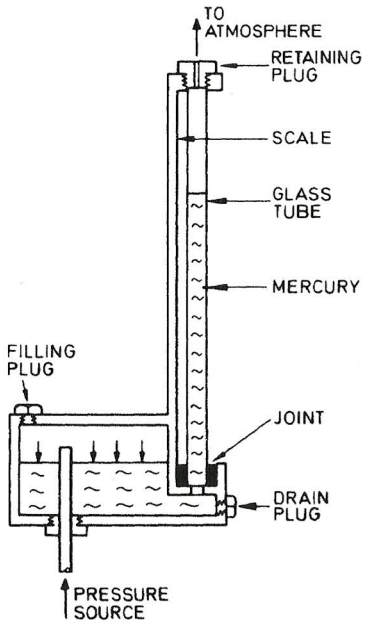


Fig. 2.2 MERCURY MANOMETER (WELL TYPE)

This instrument measures pressures of a higher order than that measured by the water manometer, such as scavenge or super-charge air pressure for IC engines. The uniform bore glass tube is small in diameter so that when mercury is displaced from the well into the tube, the fall in level of the mercury in the well is so small it can be neglected. Hence the pressure reading is indicted directly by the level of the mercury in the glass tube. The relative density of mercury is 13·6 hence 1 mm of mercury is equivalent to a pressure of 9·81 x 13·6 N/m^2, *i.e.* 134N/m^2 or 0·134kN/m^2. A special application is the vacuum gauge (kenotometer) which is a combined barometer and manometer with the scale on the right hand side calibrated in absolute pressure.

MERCURY BAROMETER

A sketch would be similar to Fig. 2.2 but with the top of the glass tube sealed at a vacuum and the pressure source would be the atmosphere.

If we assume the atmospheric pressure is supporting a column of mercury 760 mm in the tube, then:

atmospheric pressure = 760 x 0·134
= 102 kN/m^2
=1·02 x 10^5 N/m^2
=1·02 bar

For an atmospheric pressure of 760 mm the equivalent water barometer would be 760 x 13·6, *i.e.* 10 336 mm or 10·336 m.

The value of the atmospheric pressure varies slightly with climatic conditions hence to ascertain true absolute pressures the barometer reading should be taken at the same time as a gauge pressure is taken.

If for example we wish to obtain the absolute pressure in a condenser and the readings were:

condenser gauge reading 742 mm

barometer gauge reading 762 mm

then, condenser pressure = (762 – 742) x 0·134
= 2·68 kN/m^2

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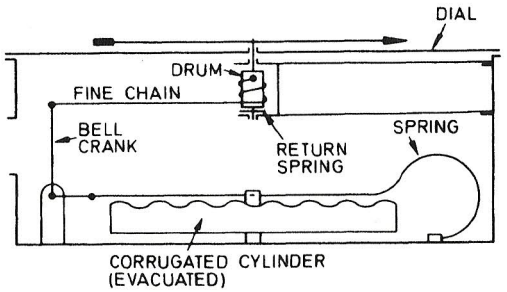

Fig. 2.3 ANEROID BAROMETER

ANEROID BAROMETER

The aneroid barometer is shown in Fig. 2.3. It consists of a corrugated cylinder (detecting element) made of phosphor bronze or other similar material, a steel spring, bell crank, pointer, dial and case (measuring element). The corrugated cylinder is completely evacuated hence the pressure of the atmosphere tends to collapse it. The centre of the corrugated cylinder deflects downwards if atmospheric pressure increases and the spring causes deflection upwards if atmospheric pressure decreases. Cylinder motion is transmitted to the instrument pointer.

PRESSURE GAUGE (BOURDON)

A pressure relay tube is the principal working component (detecting element). This tube which is semi-elliptical in cross section is connected to the pressure source. When the tube is subjected to a pressure increase it tends to unwind or straighten out and the motion is transmitted to the gauge pointer through the linkage, quadrant and gear (measuring element). If the tube is subjected to a pressure decrease it winds, or coils, up and the motion is again transmitted to the pointer. This gauge is therefore suitable for measuring pressures above or below atmospheric pressure. A diagrammatic sketch is shown in Fig. 2.4.

Materials used in the construction of the gauge are solid drawn phosphor bronze or stainless steel for the pressure relay tube. Bronze or stainless steel for the quadrant, gear and linkage. Case, brass or plastic.

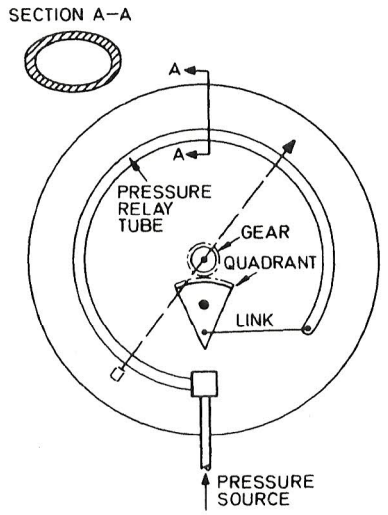

Fig. 2.4 PRESSURE (OR VACUUM) GAUGE – BOURDON

The Bourdon movement is frequently used in transducers and controllers to vary output signals in pneumatic or electrical form.

PRESSURE GAUGE (SCHAFFER)

This type utilises a strong flexible metal diaphragm (detecting element) which moves up as pressure increases. The device is shown in Fig. 2.5. Again this device can also be used as a transducer (pneumatic or electric) in telemetering or control with an output signal proportional to diaphragm movement. Similar remarks apply to most detecting (sensing) devices (as detailed in Chapter 6).

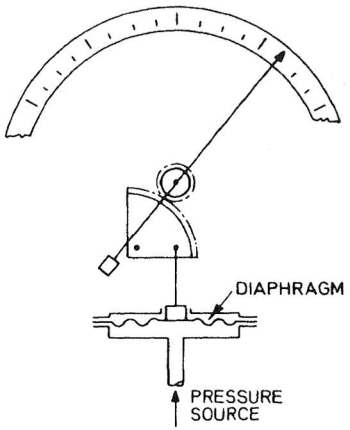

Fig. 2.5 PRESSURE (OR VACUUM) GAUGE – SCHAFFER

DIFFERENTIAL PRESSURE CELL (D/P CELL)

The sketch of Fig. 2.5 illustrates a single diaphragm subject to differential pressure. The d/p cell is often used in direct differential pressure recording as well as flow and level applications. The detecting element of the cell is a bellows, or diaphragm, whose mechanical movement is used to indicate or transduced to electrical or pneumatic signal output. Low pressure - slack diaphragm.

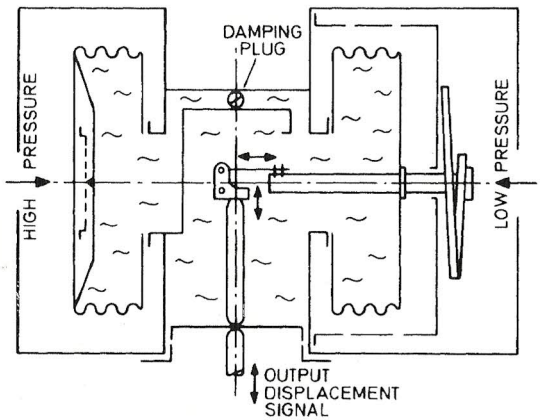

Fig. 2.6 DIFFERENTIAL PRESSURE CELL

As an alternative to a one membrane diaphragm a sealed capsule (twin membrane) can be inserted in the cell body and secured so that different pressures are applied at each side. The capsule can be filled with a constant viscosity fluid (for a fairly wide temperature range) which also damps oscillation. Silicone is such a fluid. Again mechanical movement of the capsule is proportional to differential pressure. Capsule stacks are also used.

Another type of d/p cell utilises two separate bellows. Such a design, often called a Barton cell, is shown in Fig. 2.6. Refer now to this sketch.

Pressure increase on the high pressure side displaces liquid (via an adjustable damping plug) and expands the right hand low pressure bellows. This bellows is connected to the horizontal spindle, one end of which has a flexure strip and the other end is fastened to flat plate springs. Equilibrium exists when spring force equals differential pressure. Mechanical travel is via the flexure strip, lower seal diaphragm and vertical spindles to indicator or transducer. A bi-metallic strip adjusts bellows fluid capacity to allow for volumetric expansion. This device is shown dotted within the left hand bellows. The bellows are often made of welded stainless steel discs and inner faces butt if excess pressure is applied, so protecting the bellows.

PIEZOELECTRIC DETECTING ELEMENT (SENSOR)

With *certain* solid crystals having an asymmetrical electric charge distribution, any deformation of the crystal produces equal external unlike electric charges on the opposite faces of the crystal. This crystal is known as the piezoelectric effect.

Deformation of the crystal can be caused by pressure and the charges produced can be measured by means of electrodes attached to the opposite surfaces of the crystal.

Crystal materials can be naturally occurring or man made. Quartz (*i.e.* $Si0_2$) is a material that can be used in temperature environment up to 550°C, it is stable mechanically and thermally. Barium titanate, a ceramic produced commercially, is also used but tourmaline is principally used because of its good electrical properties. The output voltage from tourmaline crystal, acting as a detecting element (sensor) and transducer, is a linear function of the pressure applied and the charge sensitivity is approximately 2

x 10^{-10} coulomb/bar, it can be used for pressures varying from 1·03 to 800 bar. The only drawbacks are (1) its sensitivity to temperature change (*i.e.* its pyroelectric effect) hence ideally it should only be used under controlled temperature conditions (2) it is more suited to pressure variations than to static pressure.

Figure 2.7 shows crystals in series and parallel. The series arrangement gives a higher output voltage for the same pressure applied whereas the parallel arrangement reduces output impedance and hence is a more stable system. Figure 2.8 is a commercial piezoelectric sensor using tourmaline crystals in parallel, the sensor is small and is coated with insulating material to help minimise the effect of temperature variation in the medium whose pressure is being measured.

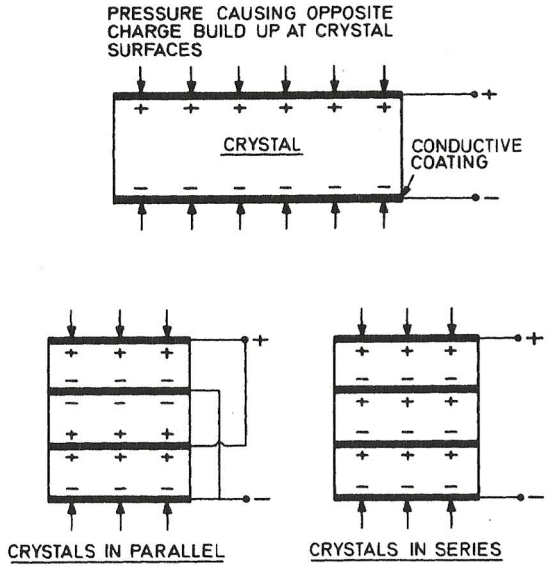

Fig. 2.7 PIEZOELECTRIC CRYSTALS

Pyroelectricity, like piezoelectricity, makes use of the spontaneous polarisation of materials such as ferro-electrics. Temperature variations causing changes in polarisation can be utilised in very accurate calorimetric recordings.

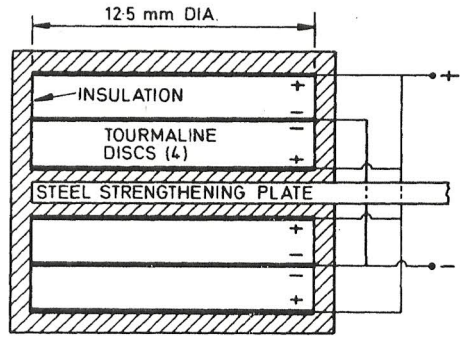

Fig. 2.8 PIEZOELECTRIC SENSOR

STRAIN GAUGE

A strain gauge is fundamentally a resistive wire of about 0·01 mm diameter subject to strain by pressure (force or acceleration) with electrical resistance change proportional to strain. *Bonded* elements of the wire wound type are either flat grid or helical wound on a former. The wire is fixed to a backing material such as paper, resin or plastic which is glued to the surface under test, wires are soldered or spot welded. One alternative is the foil type where the grid is etched from thin metal foil using printed circuit techniques. Another alternative is a *p* or *n* doped silicon semi-conductor which can be very small and is extremely sensitive.

The *unbonded* strain gauge is essentially a pressure sensor and a typical design is shown in Fig. 2.9.

The detecting (sensing) element can be fastened directly to the diaphragm as shown, or alternatively, a central force rod can transmit diaphragm movement to the detecting (sensing) element consisting of plate springs with posts on the periphery upon which the strain gauge wire is wound.

In all strain gauges, to minimise resistance change due to temperature effects, it is usual to employ materials with a low temperature coefficient of resistance for the wire, such as constantan. Alternatively a second compensating wire loop can be incorporated.

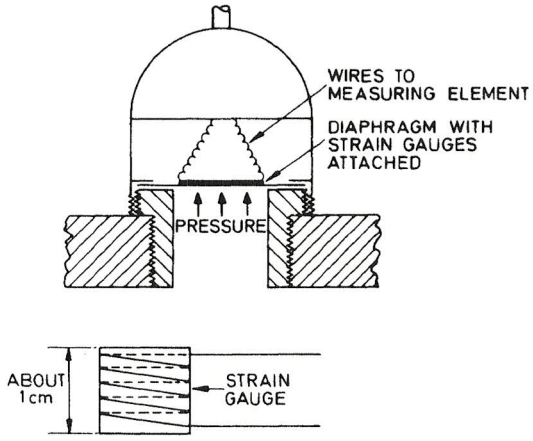

Fig. 2.9 STRAIN GAUGE PRESSURE SENSOR

The measuring element for strain gauges is generally a Wheatstone bridge circuit with temperature compensating resistance and strain gauge resistance arranged as two, of the four, resistances and a central galvanometer and constant dc voltage source. Null balance methods can also be used, or for high sensitivity the bridge may be fed by ac voltage and the galvanometer replaced by a transistor amplifier and detector. The measuring unit is effectively a transducer, *i.e.* displacement-electrical..

TEST EXAMPLES 2

1. Sketch and briefly describe three types of pressure measuring device.
2. Sketch and describe a Bourdon type of pressure gauge. State the materials used in construction. Discuss briefly how the Bourdon movement can be utilised for telemetering devices.
3. Describe, with the aid of a sketch, any type of differential pressure cell. Detail three applications in instrumentation of the use of the d/p cell.
4. Explain the operation of a foil strain gauge and describe a bridge method to achieve an output voltage proportional to the strain.

CHAPTER 3

LEVEL MEASUREMENT

Liquid level sensors are usually classified under two headings, i.e. direct methods and inferential methods.

DIRECT METHODS

FLOAT OPERATED

The float is generally a hollow cylinder or ball working on direct action or displacement principles. Level variation is transmitted by chain or wire and pulley or torque tube (usually with counterweights fitted) to the indicator. High or low level alarm contacts are easily arranged. Pulley movement can also be arranged to operate a contact arm over an electrical resistance so varying current or voltage to indicator or receiver.

SIGHT GLASSES

Various types are in use dependent on working conditions. The simple boiler water glass gauge with toughened glass and the plate type of water gauge for high pressures are typical.

PROBE ELEMENTS

Floatless types of level sensors can be arranged where the liquid is a conductor. Sensing electrodes, rods or discs, vary electrical circuits when they are in contact with liquids. A typical example is detection of the fluid level in a tank by capacitive techniques (see Fig. 3.1). This technique is used in oil-water interface detection on oily-water separators - see Chapter 13.

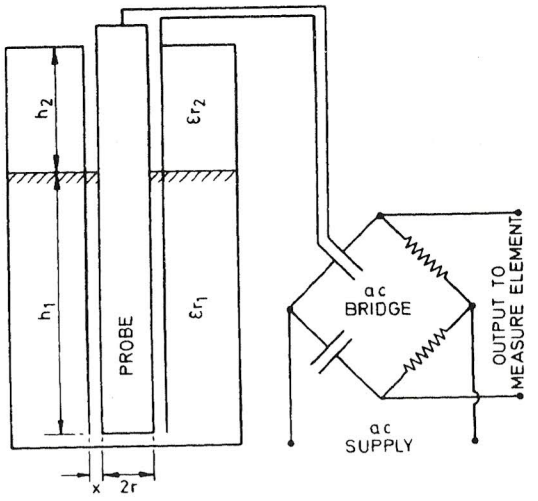

Fig. 3.1 LEVEL SENSOR (CAPACITIVE)

Detection of level is measured by variation of capacitance which is accomplished by alteration in dielectric strength. One plate of the capacitor is a probe, possibly made of stainless steel, the other is the shell and both are connected to an ac bridge which is supplied with high frequency low voltage alternating current. As the interface moves the dielectric strength (relative permittivity ε_r) alters.

An approximate expression for the capacitance in an ac bridge circuit (Fig. 3.1) is:

$$C = \frac{\varepsilon_0[\varepsilon_{r1}h_1 + \varepsilon_{r2}h_2]}{2l_n[1 + x/r]}$$

and since ε_{r2} is approximately unity for air ($\varepsilon_{r1} > 1$)

$$C = \frac{\varepsilon_0[\varepsilon_{r1}h_1 + h_2]}{2l_n[1 + x/r]}$$

where ε_0 is the free space permittivity, ε_r relative permittivity
" h_1 " " head of fluid being measured
" h_2 " " distance from fluid surface to tank top
" $2r$ " " diameter of the probe
" x " " separation between the circular plates
" l_n " " natural logarithm.

INFERENTIAL METHODS

PRESSURE ELEMENTS

The static-pressure method is extensively used.

$$p = \rho g h$$

p is pressure, ρ density of fluid, g gravitational acceleration, h fluid head. Any of the sensing and measuring devices described in Chapter 2 are therefore applicable to level measurement. In particular a pressure gauge, calibrated in height units, is probably the most simple indicator. Bellows or diaphragms are used with the pressure bulb located inside or outside the vessel with suitable correction factors applied for correct datum and density of fluid. The differential pressure cell is often used in level measurement. Telemetering for remote reading is readily applied, pneumatically or electrically, to the displacement movement of the level (pressure) sensor.

MANOMETER TYPES

The static pressure equation is applicable and devices have been described in Chapter 2. Three types of remote level indicator are now described.

For the Electroflo electrical type as sketched in Fig. 3.2 the difference in level h is directly proportional to the difference between the level of the liquid in the tank and a datum. The transducer element contains resistances immersed in mercury to form an electrical circuit with transmission to remote indicator receivers.

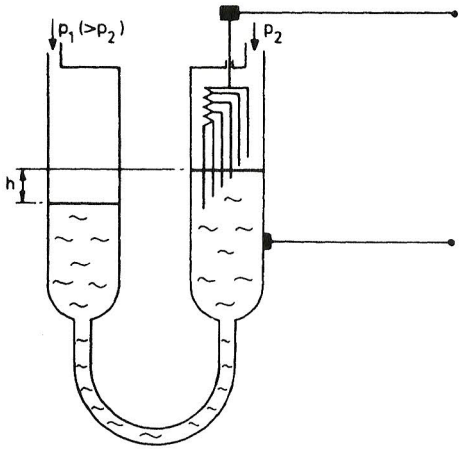

Fig. 3.2 LEVEL SENSOR/TRANSDUCER (RESISTIVE)

Figure 3.3 is a diagrammatic arrangement of the Igema remote water level indicator. The lower portion of the U tube contains a (red) coloured indicating fluid which does not mix with water and has a density greater than that of water.

The equilibrium condition for the gauge is $H = h + \rho x$ where ρ is the relative density of the indicating fluid. H, h and x are variables.

If the water level in the boiler falls, h will be reduced, x will be increased and H must therefore be increased. The level of the water in the condenser reservoir being maintained by condensing steam.

If the water level in the boiler rises, h will be increased, x will be reduced and H must therefore be reduced. Water will therefore flow over the weir in the condenser reservoir in order to maintain the level constant.

A strip light is fitted behind the gauge which increases the brightness of the (red) indicating fluid, which enables the operator to observe at a glance, from a considerable distance, whether the gauge is full or empty.

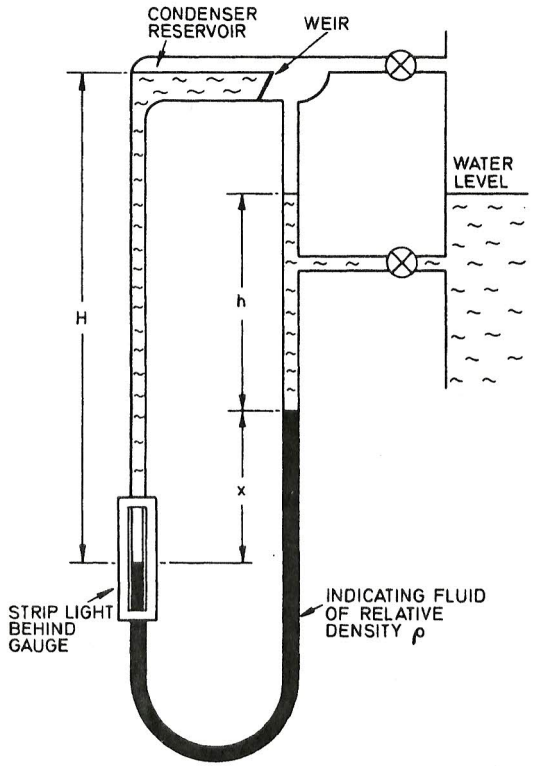

Fig. 3.3 IGEMA REMOTE WATER LEVEL INDICATOR

Figure 3.4 is another type of remote water level indicator. In this case the operating fluid is the boiler water itself. The operation of the gauge is as follows:

If we consider a falling water level in the boiler, the pressure difference across the diaphragm h will increase, causing the diaphragm to deflect downwards. This motion of the diaphragm is transmitted by means of a linkage arrangement (see insert) to the shutter which in turn moves down pivoting about its hinge, causing an increase in the amount of (red) colour and a decrease in the amount of (blue) colour seen at the glass gauge.

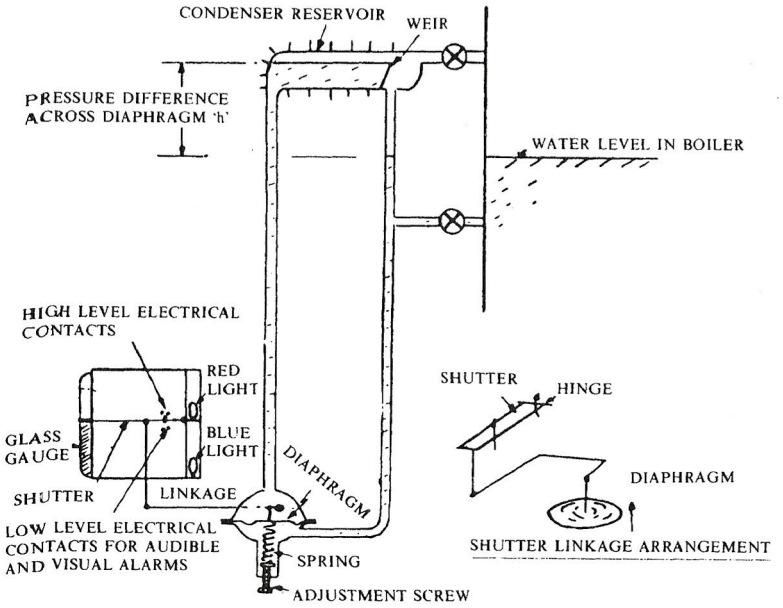

Fig. 3.4 REMOTE WATER LEVEL INDICATOR

It will be clearly understood that if the water level now rises then the (red) will be reduced and the (blue) increased.

Separating the (blue) and (red) colours, which are distinctive and can clearly be seen from a considerable distance, is a loose fitting black band which moves with the shutter, giving a distinct separation of the two colours.

An adjustment screw and spring are provided to enable the difference in diaphragm load to be adjusted. Hence correct positioning of the shutter and band in relation to the reading of a glass water gauge fitted directly to the boiler is possible.

Both devices shown in Figs 3.3 and 3.4 are also capable of being observed by closed circuit television systems to extend the distance between transmission and reception. Alternatively telemetering by pneumatic or electrical transmitters is readily arranged. A d/p cell would be very suitable for the former and any displacement transducer, such as a differential transformer, could be used for the latter.

PURGE SYSTEMS

For small air flow rate, about one bubble per second, a pressure equal to that in the dip tube will be applied to the indicator as shown in Fig. 3.5. This simple bubbler device is an arrangement that is similar to the well known **pneumercator** used for determining depths of water and oil in tanks. Air supply to the open ended pipe in the tank will have a pressure which is directly proportional to the depth of liquid in the tank.

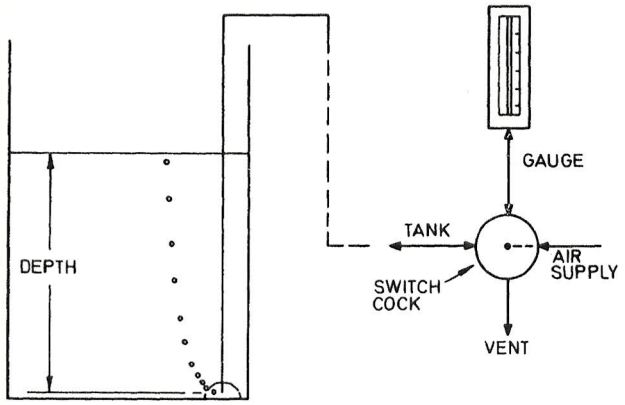

Fig. 3.5 PNEUMERCATOR LEVEL INDICATOR

ULTRASONIC AND NUCLEONIC DEVICES

An *ultrasonic* transmitter and receiver (utilising piezo-electric crystals) are located above the tank so that two echoes are received, one from the liquid surface and one from the tank bottom. The time separation between the two signal echoes is proportional to liquid depth - which can be displayed and measured by a cathode ray oscilloscope (CRO).

Nucleonic units have a shielded radioactive strip source from which gamma radiation is picked up by a detector on the opposite side of the storage vessel.

TEST EXAMPLES 3

1. Sketch and describe a distant-reading boiler water level gauge. Explain its principle of operation.
Give two ways in which the gauge would give incorrect readings.
State the routine maintenance necessary to ensure maximum reliability at all times.

2. Sketch and describe any type of level sensor. Discuss the modifications necessary to utilise the device as a transmitter in a telemetering system.

3. Describe, with a sketch, an instrument used to indicate the level of liquid in a tank. Can the result be used to determine pressure above a given datum and, if so, state the other variables likely to be involved.

4. Describe the arrangement of an air purge tank contents system.

CHAPTER 4

FLOW MEASUREMENT

Many of the techniques utilised in flow measurement employ principles also used in pressure and level measurements.

BASIC PRINCIPLES

Flowmeters are generally divided into two fundamental types, *i.e.* quantity meters and rate of flow meters, use of the word flowmeter generally implies the latter.

QUANTITY METERS

These devices measure the *quantity* of fluid that has passed a certain point. *No time is involved.* Types are usually classified as *positive* or *semi-positive*. A typical positive type utilises the flow to drive a reciprocating piston and a counter is attached. The meter acts like a conventional engine with fluid pressure supplying motive power. Stroke length and cylinder dimensions fix the quantity delivered per cycle. Semi-positive types are usually rotary. A form of gear pump, or eccentrically constrained rotor, can be used which is driven by the fluid. Quantity is measured by number of rotations (cycles) and fluid per cycle.

RATE OF FLOW METERS

Measure the velocity of fluid passing a certain point at a given instant. From this rate of flow (quantity per unit time) can be determined from velocity multiplied by area of passage. They are therefore classified as *inferential, i.e.* volume inferred from velocity. There are two fundamental components of the rate of

flow meter. The *primary* element is that portion of the instrument which coverts the quantity being measured into a variable to operate the secondary element, for example, orifice and pressure tappings from a venturi. The *secondary* element measures the variable created by the primary element, for example, a differential pressure cell.

INTEGRATORS

Quantity meters are more expensive and less suited to deal with large fluid quantities than rate of flow meters. Rate of flow meters are often used as quantity meters by fitting an integrator. As a simple example consider variable flow. The rate of flow can be measured at set time intervals and a graph plotted, the area of which gives quantity over the time period required. In practice this is performed mechanically or electrically by an integrator. One type is based on the planimeter principle and another type (escapement) utilises mid-ordinate techniques from a heart-shaped cam drive. Flat faced cam drive, or worm and wheel designs, can be used with a turbine wheel or helix type of primary element having a counting mechanism secondary element, incorporating the integrator to interpret quantity. The integrator is often included within the receiving unit of telemetering systems and a typical device is illustrated and described in Chapter 6. Integration is readily performed electrically by use of a conventional watt-hour meter. It should be understood that integration is a general instrumentation operation whose use is not restricted only to flowmeters.

SQUARE ROOT EXTRACTION

When inferential devices are used, with velocity sensors utilising differential pressure techniques, the velocity is not directly proportional to pressure difference, or head. Velocity is related to the square root of pressure, or head, i.e. a curve of flow rate plotted against pressure, or head, is of parabolic form.

This means that if a pressure difference is used in a sensor device connected to a manometer, or pointer through a linear mechanism, the rate of flow scale on the manometer or traversed by the pointer would have to be a square root function. The scale divisions would increase in square root increments for equal

increments of flow rate. From the aspect of display and continuous recording of flow rate the square root characteristic is not an embarrassment. If however the differential pressure has to be used in a control system the square root is usually extracted to give a signal which is directly proportional to the fluid flow rate. Square root extraction is described later in this chapter under differential pressure inferential devices. It is difficult to integrate readings of a system when the square root extraction is not applied.

INFERENTIAL-ROTATIONAL

Basically these can be considered as mechanical or electrical.

MECHANICAL TYPE FLOWMETER

Designs are usually of the "turbine wheel" type with speed of rotation directly proportional to linear flow velocity and, with area of passage fixed, the volume rate is inferred. A wheel, fan or helix is inserted in the pipe or duct, mounted vertically or horizontally, gear trains are used to interpret the movement. This is the principle of vane anemometers.

Rotormeter

The measuring principle is illustrated in Fig. 4.1 where the meter, of the duplex rotor positive displacement type, is shown in three specific positions. This rotary flowmeter operates on the displacement principle, the measuring system consisting of a casing with two rotors. Bearing bushes are provided on either side of each rotor so that the rotor runs clear of the casing and spindle in a radial direction. The rotor is located in an axial direction by means of end bearings so that it runs clear of the bearing plates. Both the end bearings and the spindle are fixed to the shaft whilst the bearing bushes are fixed to the rotor and rotate relative to the end bearings and the shaft. Each rotor carries a gear wheel at the rear through which the rotors are coupled together. The rotary movement of the rotors is transmitted to the outgoing shaft of the

meter through a gearwheel fitted at the front of one of the rotor. Consider Fig. 4.1:

In position "A", the left-hand rotor is fully relieved from load whilst the liquid pressure acts on one side of the right-hand rotor

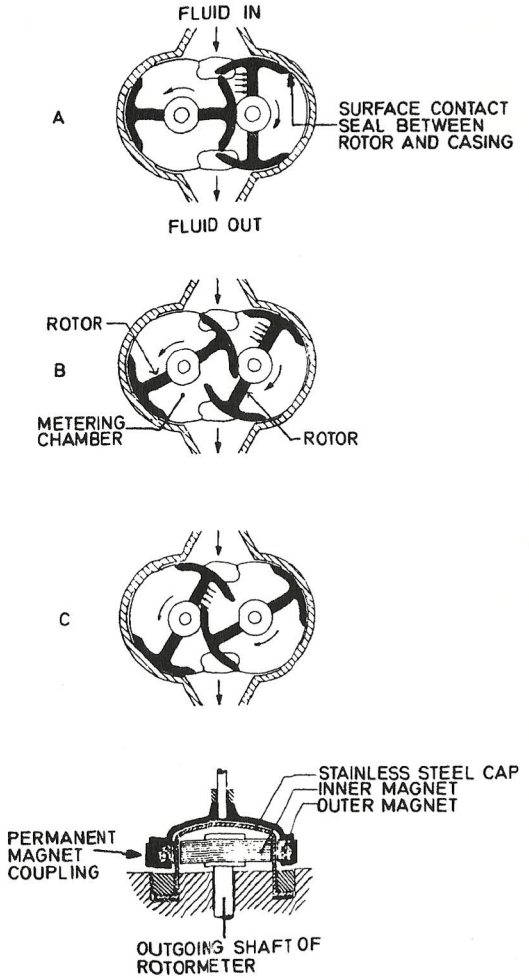

Fig. 4.1 FLOWMETER (MECHANICAL)

causing this rotor to rotate in clockwise direction. Since both rotors are coupled together through gears, the left-hand rotor will rotate in an anti-clockwise direction.

In position "B", the liquid in the displacement chamber is pressed by the right-hand rotor to the outlet.

In position "C", the right-hand rotor is entirely relieved from load whilst the liquid pressure now acts on one side of the left-hand rotor so that it takes over the task of the right-hand rotor.

To reduce leakage losses to a minimum, it is essential to provide for effective sealing between inlet and outlet. To this end, the rotors seal off against the casing by surface contact.

In the rotormeter, the motion of the rotors is transmitted to the external parts by the attraction between two permanent magnets, an inner and outer magnet. The maximum torque transmitted by this system is 0·4 N m. This arrangement offers the following practical advantages;

(1) Perfectly leakproof transmission, which means that it is impossible for corrosive or hazardous liquids to leave the meter.

(2) Protection of attached parts or instruments. If the mechanism of external parts or instruments should be blocked for whatever reason, the permanent magnetic coupling will slip, thus avoiding any damage to the instruments.

Although the torque to be transmitted by the magnets may vary with the external parts or the type of instrument used, it will hardly ever exceed a value of 0·05 N m with the meter running at constant speed. Hence, at ambient temperature the magnetic coupling has an 8-fold safety margin which will, in most instances, be amply sufficient to take accelerations and decelerations (which are the most frequent operating conditions) without slipping of the coupling.

Although the torque transmitted by the magnets decreases with rising temperature, experience has shown that even with a liquid temperature of up to 250°C the available torque is still amply sufficient.

ELECTRICAL TYPE FLOWMETER

One type utilises rotating vanes with a small magnet attached to one vane and a coil in the pipe wall. The electrical impulse can be

counted on a digital tachometer calibrated in flow rate. The design now described has no moving parts.

Electro-magnetic flowmeter

This type is shown in Fig. 4.2.

The principle utilised is that of a moving conductor (the liquid) in a magnetic field generating a potential difference. In the simple arrangement shown the two electro-magnets are supplied with current (ac is preferred to dc to reduce polarisation of the dielectric). There are two sensor electrodes. If B is the flux density of the field, v velocity of flow, d pipe diameter, then in suitable units the emf generated at any instant is given by:

$$e = Bvd$$

For constant B and d, e is directly proportional to v.

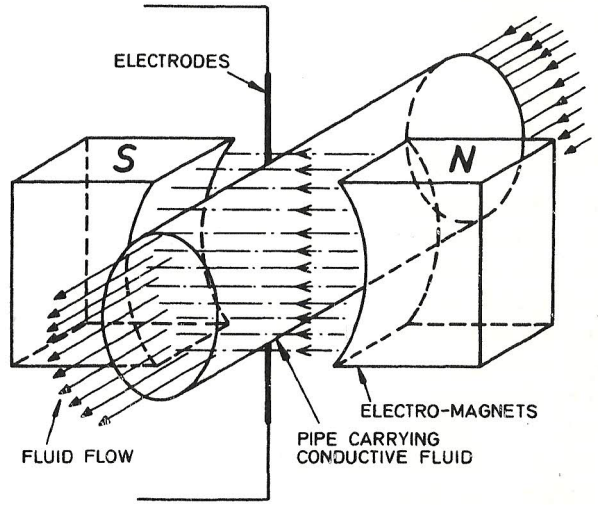

Fig. 4.2 FLOWMETER (ELECTRICAL)

ROTAMETER

This type does not strictly fit into the classifications given but a brief description is appropriate at this stage. A sketch is given in Fig. 4.3.

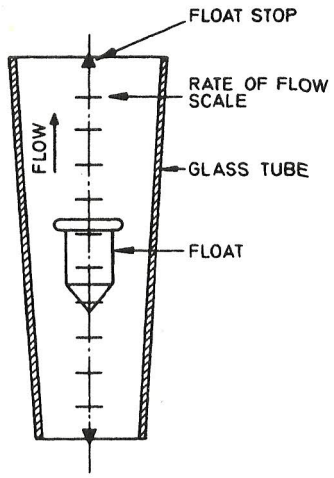

Fig. 4.3 ROTAMETER

This is a variable area meter. A long taper tube is graduated on its vertical axis. The float moves freely in the tube and by an arrangement of shaped flutes in the float it slowly rotates. As flow rate increases the float rises in the tube, so that the annular area increases, which means that the differential pressure across the tube is at a constant value. The float can be arranged with a magnet attachment and a follower magnet outside will transmit motion to a pointer via linkage if required.

INFERENTIAL-DIFFERENTIAL PRESSURE

PRIMARY ELEMENTS

The orifice and the venturi will be described and these are mainly used although flow nozzles and special inserts, such as a Dall tube, are also employed. Fig. 4.4 shows both the orifice plate and venturi sensors using energy conversion to produce a pressure difference which can be utilised by the secondary element to provide a signal for direct reading, telemetering or control.

Using the venturi flow sensor as an example the theory involved is as follows:

Assuming unit mass, and energy at points 1 and 2 being the

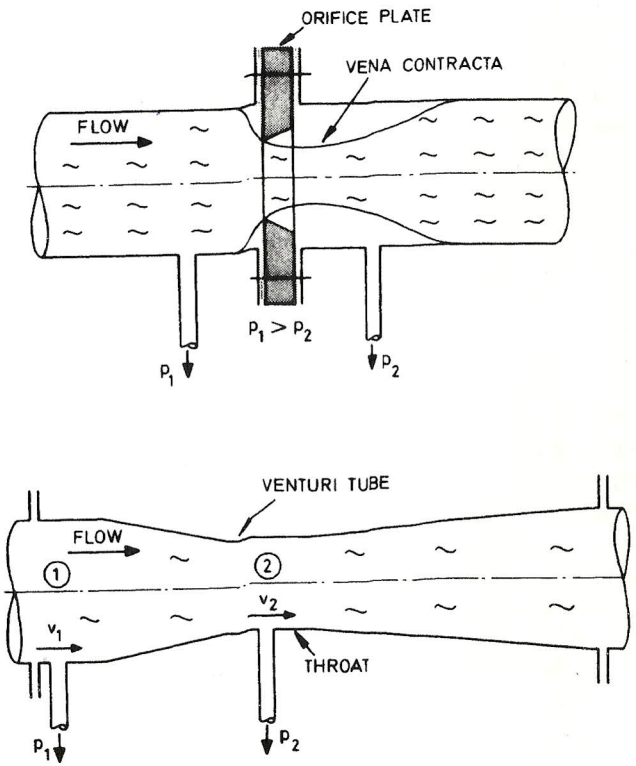

Fig. 4.4 FLOW SENSORS

same, i.e. neglecting friction and shock losses as small, then from Bernoulli for incompressible flow of fluid of density ρ:

$$\text{KE at 1} + \text{PE at 1} = \text{KE at 2} + \text{PE at 2}$$

$$\tfrac{1}{2}v_1^2 + p_1/\rho = \tfrac{1}{2}v_2^2 + p_2/\rho$$

where KE is kinetic and PE is pressure energy.

The equation for continuity of flow for area A is;

$$v_1A_1 = v_2A_2$$

By substituting for v_2 from (b) in (a) and using mass flow rate $\dot{m}$ as equal to ρv_1A_1 it can be shown that

$$\dot{m} = k\sqrt{p}$$

where p is the pressure difference $(p_1 - p_2)$ and k is a meter constant in terms of areas and density which includes a discharge coefficient factor. Frictional losses are greater for orifice than for venturi meters. Seal pots protect sensor leads.

SECONDARY ELEMENTS

Any differential pressure device can be used as a secondary element including the manometer, diaphragm and d/p cell as described in Chapter 2. The measure scale will be non-linear for direct recorders due to the square root relation and telemetering, control and integration will be generally unsatisfactory unless a correcting unit is fitted. When manometers are used various compensations can be used. The simple manometer can utilise a curved measuring limb and the well type manometer can be arranged with a shaped chamber or may include a parallel tube and a shaped displacer. Other direct measuring devices utilise a cam incorporated in the mechanism, an example is the ring balance. In the electrical resistive sensor described in Chapter 3 (Fig. 3.2), when used for flow measurement, the electrode tips immersed in mercury are arranged in a parabolic curve with each other which gives the compensation. Three flow sensor square root extraction devices are now considered in more detail.

SQUARE ROOT EXTRACTORS

Figure 4.5 is one type of square root extractor using a parabola shaped bell which can be connected through linkage to mechanical, pneumatic or electrical display and control systems.

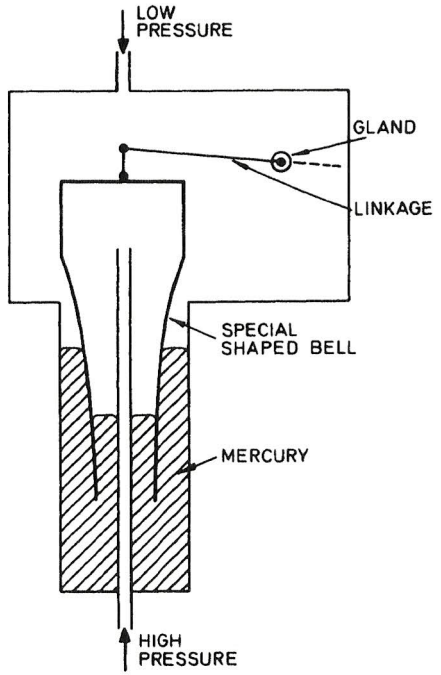

Fig. 4.5 SQUARE ROOT EXTRACTOR (MECHANICAL)

The differential pressure is applied with the high pressure inside and the low pressure outside the bell.

With its changing cross sectional area and buoyancy, due to change in differential pressure, the bell movement is made to be directly proportional to the square root of the differential pressure. Hence the bell movement is directly proportional to the fluid flow being sensed by the orifice plate or venturi sensor.

This device is often called a Ledoux Bell. For air flow measurement, instead of the bell being shaped. a shaped displacer is arranged in a separate chamber. The displacer is connected to rise with the bell. Such a device is used for measuring steam and air flow, and for controlling steam/air flow ratio, the characteristic is linear.

Figure 4.6 is a diagrammatic sketch of another type of square root extractor utilising the pneumatic flapper nozzle position balance principle (as described in Chapter 6).

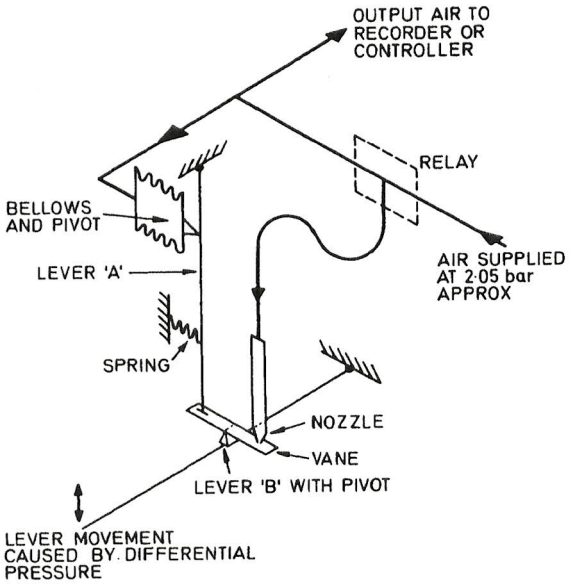

Fig. 4.6 SQUARE ROOT EXTRACTOR (PNEUMATIC)

Refer to Fig. 4.6:

The differential pressure from a flow sensor acts on a horizontal lever B, this effects the amount of air escaping from the nozzle and hence the pressure in the bellows. Pressure alteration in the bellows causes movement of the vertical lever A. The very small relative motion between levers A and B provides square root extraction. The output air signal is directed to the measure element (recorder) or controller.

Fig. 4.7 shows a square root extraction technique using electrical force balance (as described in Chapter 6). The differential

pressure, in electrical signal form represented as variable x input, is applied to the left hand side. With the force balance beam in equilibrium the output signal is variable $\sqrt{x}$. Variation in input signal, causing unbalance, can be arranged to be re-balanced by suitable adjustment to output signal. This can be done in various ways, one method could be to connect the right hand of the beam to a differential capacitor in the output circuit (see Fig. 6.6). This is a closed loop but without electrical connection between input detected signal and output measured signal. The force balance system is a simple analogue computer - output can also be easily arranged for squaring, summing, etc, by suitable wiring combinations.

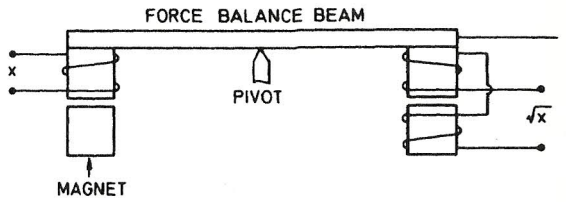

Fig. 4.7 SQUARE ROOT EXTRACTOR (ELECTRICAL)

Figure 4.8 shows two arrangements of flow sensor/transducer units, each with square root extraction. For the pneumatic system the valves marked A, B and C would be used in the sequence, open B, close A and C, when taking the differential pressure transmitter out of operation. This valve sequence would also apply to the electric system.

With steam flow measurement, the pressure tappings from the flow sensor are led to cooling reservoirs wherein the steam condenses. Water only then acts on the square root eliminator or recorder thus preventing damage.

The pneumatic system would utilise square root extraction as shown in Fig. 4.6 (the technique shown in Fig. 4.5 could also be used).

The electrical system can utilise mechanical movement (from a device such as Fig. 4.5) as shown to give square root extraction. Variation of the position of the soft iron core of the inductor unit

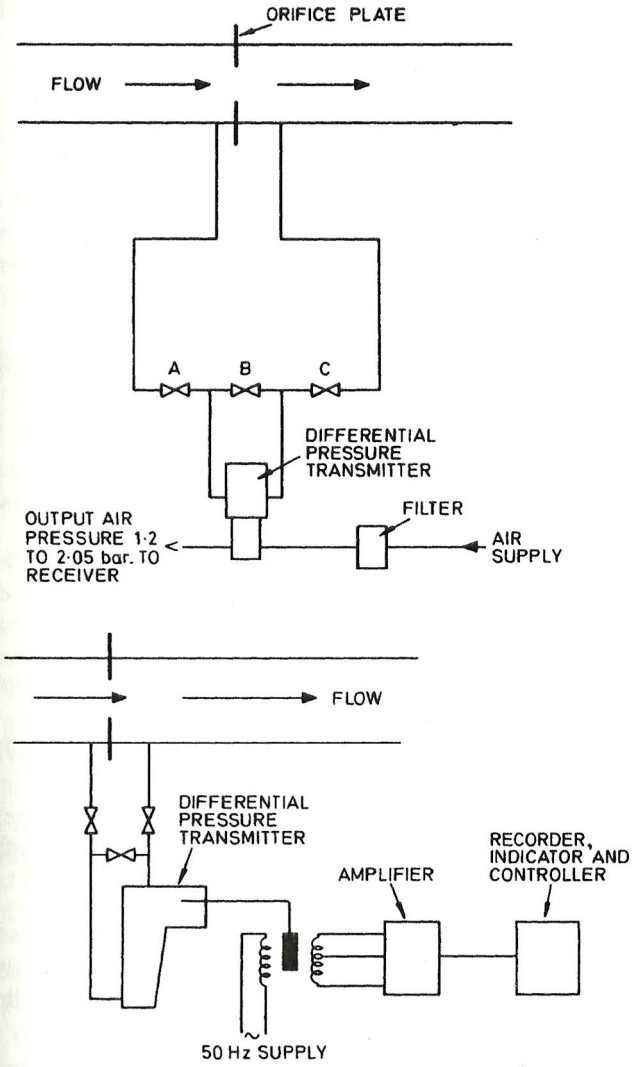

Fig. 4.8 FLOW SENSORS/TRANSDUCERS

govern output signal to the amplifier. Alternatively direct signals to the amplifier can be taken from electrical devices shown in Fig. 3.2 and Fig. 4.7.

ULTRASONIC AND NUCLEONIC

Ultrasonic flowmeters utilise an oscillator transmitter and receiver timer to measure apparent sound velocity in the fluid, which is proportional to the velocity of the fluid, across a diametral path inclined at an angle with the flow direction. Calibration is required to allow for the velocity profile.

Nuclear types employ a radioactive tracer injected into the fluid to measure the flow time, for a known volume, between two points by two detectors and a recorder. Normal protection against ionising radiation is required for the sample measure.

TEST EXAMPLES 4

1. Sketch the apparatus and describe the method of measuring steam flow through a pipe.
Explain why results obtained using this apparatus may differ from theoretical results.

2. Describe, utilising a sketch, any type of flowmeter. For inferential types of device how is the measured variable related to flow rate?

3. Inferential flowmeter devices using differential pressure techniques for velocity sensing exhibit non-linear characteristics. Discuss these characteristics and describe a device to restore linear characteristic to measure or control output signal.

4. Sketch an orifice plate installation showing the flow pattern and pressure variation. To what is the flow rate proportional?

CHAPTER 5

OTHER MEASUREMENTS

From an engineering viewpoint the variables discussed in the preceding chapters are the most significant. However, other variables have a particular specialist interest and a selected number are now presented with the appropriate instrumentation measurement.

ELECTRICAL

Such instrumentation covers a wide field, generally well described in electrical text books, and on this basis is not considered in this text unless specifically necessary. The measurement of capacitance, current, voltage, resistance, frequency, phase, energy, etc. are all very important in instrumentation. The reader will be familiar with such as moving iron and moving coil meters, etc. The bridge (ac or dc) and potentiometer are frequently used in telemetering and control components. Electronic measuring devices are being increasingly used and rectifiers, amplifiers, oscillators, etc. are an integral part of telemetering and control systems.

SPEED-TACHOGENERATOR

The dc tachogenerator is a small precision generator driven by the shaft whose rotational speed is required. Output voltage is directly proportional to speed. The tacho is best geared to run at maximum speed so giving maximum output signal and improved signal to noise ratio.

When used as a *tachometer* in measuring systems the output voltage from the tachogenerator is measured on a conventional

voltmeter calibrated in terms of rotational speed (an analogue device).

Mechanical type tachometers are based on centrifugal action, linked to produce lateral travel.

A conventional ac generator for use as a tachogenerator or tachometer is generally not satisfactory because frequency (and phase) as well as voltage amplitude are proportional to speed of rotation. In this case a "drag-cup" device is used. The rotor is a thin aluminium cup rotating around a fixed iron core. The stator is wound with two coils at right angles, one ac supply the other ac output. With the cup stationary there will be no output as the windings are at right angles. Induced emf with cup rotation, due to cutting of flux of supply winding, links with the output winding so giving a signal proportional to rotational speed; frequency and phase being that of input signal. The device can be used for rate of change detection. With dc supply at constant speed no emf is induced in the output coils but angular acceleration or deceleration induces a voltage proportional to this change in the output coils. A "velocity voltage" applied to a differentiation (rate of change) circuit (CR series) will give a voltage across the resistor which is approximately proportional to acceleration (especially with a small time constant). Alternatively as force is proportional to acceleration a simple spring accelerometer can be used. A digital tachometer (counter) is shown in Fig. 5.1.

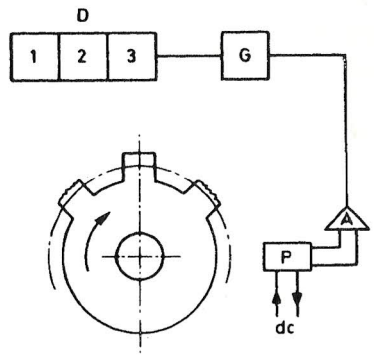

Fig. 5.1 DIGITAL TACHOMETER

As the (ferrous) toothed wheel rotates each tooth alters the air gap and flux in a pick up coil (P) whose output pulses are amplified (A). Pulses pass through a timing gate (G), say one second opening period, and are counted on a digital counter (D) which scales (related to teeth number per revolution) and displays as revolutions per second. Alternatively rev/min readings can be arranged with different gate, or scale, settings.

TORQUE-POWER

Indicated power can be measured by a conventional mechanical indicator although modern practice is tending towards oscilloscope display with integration for power. Shaft power of engines is measured by a torsionmeter in conjunction with a tachometer (power proportional to product of torque and rotational speed). Specific fuel consumption is readily achieved from these readings with a flowmeter calibration of fuel consumption. Various types of torsionmeter are available but those giving a continuous reading are usually of the electrical type. One design in common use is based on differential transformer operation (Chapter 6) which is illustrated in Fig. G of the specimen questions at the end of the book. Another design is based on magnetic stress sensitivity and is termed a *torque inductor—torductor*—and is now described.

The torductor is, as the name implies, a *torque inductor,* it is a stress transducer that is eminently suited to the measuring of torque in rotating shafts. It gives a high power output and requires no slip rings or other shaft attachments since it operates without any contact. Figure 5.2 shows a ring torductor. It consists of one primary ring which carries four poles, marked N,S, that is supplied with (50 Hz) alternating current. Two outer secondary rings have four poles each, arranged at 45° to the primaries, all of which are connected in series with mutually reversed windings.

No contact exists between the poles and the shaft, there being a 2 to 3 mm air gap provided to ensure this.

When no torque is applied to the shaft there are no stresses in the shaft and the magnetic fields between NS poles induced in the shaft will be symmetrical, the equipotential lines are then situated symmetrically under the secondary poles S_1, S_2, as shown and secondary flux and voltage will then be zero.

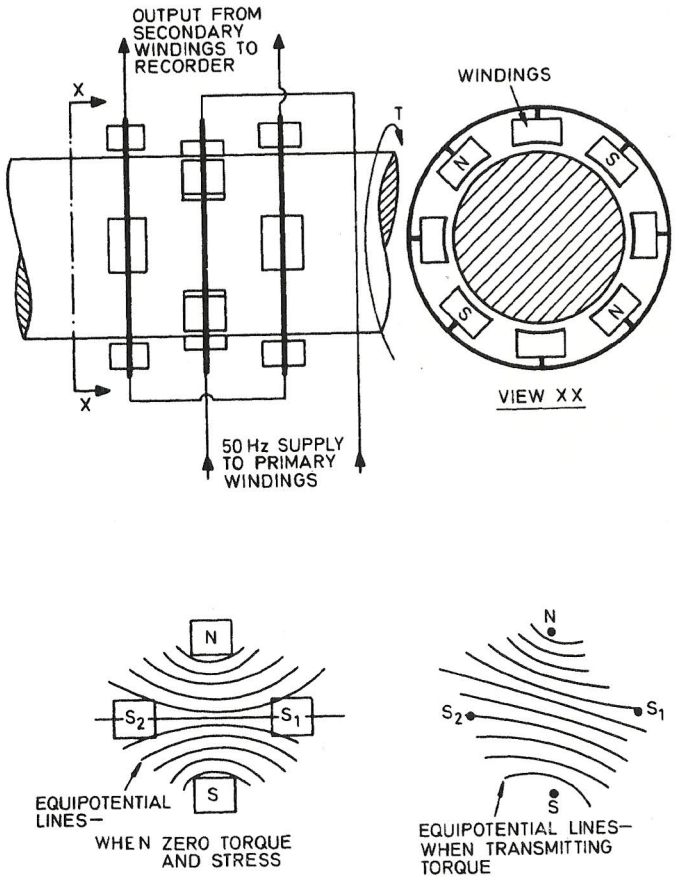

Fig. 5.2 RING TORDUCTOR

When a torque is being transmitted the equipotential lines form an asymmetrical pattern, as shown, due to the mutually perpendicular unlike stresses, acting at 45° to the shaft axis, causing increased permeability in one direction and decreased permeability at 90° in the other direction. This causes the S_1 pole to become magnetically slightly positive and the S_2 pole slightly negative.

The output from the secondaries of the ring torductor is of the order of a few milliwatts, which is large enough to be used

without any amplification. If this signal is now married to a speed signal, from a tachogenerator perhaps, then the power being developed could be displayed directly on to one dial.

VISCOMETER

Newton investigated the viscosity of fluids and postulated, for most fluids under prescribed conditions, that flow rate is proportional to applied stress, more exactly that applied shear stress is proportional to velocity gradient.

$$\frac{F}{A} = \eta \frac{dv}{dx}$$

where η is a constant called the coefficient of viscosity. Applying this equation to the case of flow through a small bore tube of radius r and length l gives:

$$\eta = \frac{\pi p r^4}{8 l \dot{V}}$$

For a constant flow rate ($\dot{V}$):

$$\eta = p \times \text{a constant}$$

where p is differential pressure.

Figure 5.3 shows the operational arrangement of the sensor element of a viscometer (see Fig. 12.12). A small gear pump driven at constant speed, by an electric motor through a reduction gear, forces a *constant* fluid quantity from the housing through a small bore tube (capillary). Fluid flows through the capillary without turbulence, *i.e.* streamline (laminar) flow prevails and pressure differential is proportional to viscosity of fluid. The pressure differential can be measured by any of the means previously described. The device, within a control system, is described in Chapter 13 (Fig. 13.19).

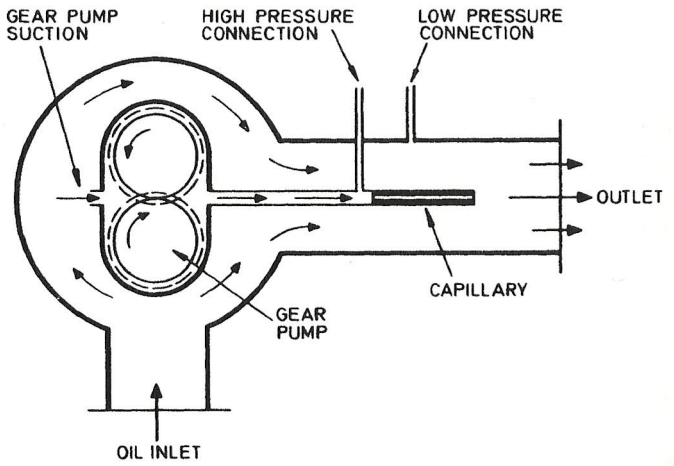

Fig. 5.3 VISCOSITY SENSOR

PHOTO-ELECTRIC CELLS

Photo-conductive cells are constructed with a thin layer of semi-conductor material and their resistance varies with the incident light energy. They are used as sensors in many situations such as oil-water content, smoke density, oil mist, flame indicator, etc. detection as described later in this chapter.

Photo-emissive cells rely on the light energy providing energy to release electrons from a metallic cathode.

If visible light, which is radiation and hence energy, falls upon certain alkali metals — such as caesium — electrons will be emitted from the surface of the metal. Metals in general exhibit this characteristic but for most materials, the light required has a threshold wavelength in the ultra violet region so that visible light does not cause electron emission.

Light energy comes in packages called *photons* and the energy of the photons is used in doing work to remove the electrons and to give the electrons kinetic energy after escape from the metal.

Figure 5.4 shows a simple photocell, visible light falls on the metal cathode from which electrons are emitted, they collect at

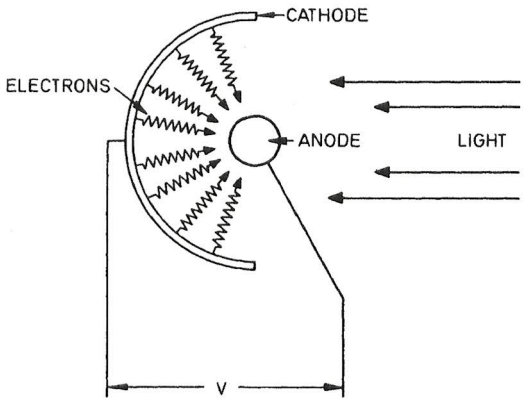

Fig. 5.4 PHOTOCELL

the anode and in this way create a potential V which can then be amplified and used for alarm and control, etc.

In the vacuum cell all current is carried by photo electrons to the positive anode. In the gas filled cell emitted electrons ionise the gas, producing further electrons, so giving amplification. Secondary-emission (photo-multiplier) cells utilise a series of increasingly positive anodes and give high amplification.

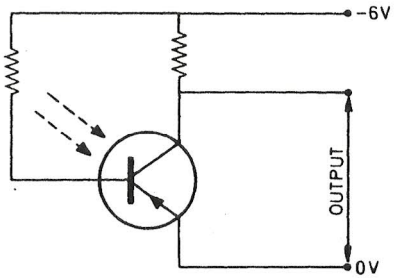

Fig. 5.5 PHOTO-TRANSISTOR

Photo-transistors exhibit similar characteristics and small size and high amplification make their use particularly attractive especially when applied to counting systems, *i.e.* digital tachometry. This device is shown in Fig. 5.5. Optical focus incident light on to the base increases the base current, hence collector current, and output voltage falls.

OIL IN WATER SENSOR

A useful application of photo-cells is in detection of oil-water interface (as an alternative to the method described in Chapters 3 and 13). Fluid passing through glass is exposed to long wavelength light from an ultra violet lamp which causes fluorescence if oil particles are present. This light can be detected by the secondary element photo-cell unit which produces a signal for amplification. The amount of fluorescent light is dependent on the amount of oil in the oil-water mixture and this affects the amount of visible light detected by the photo-cell. An ultrasonic beam between piezo-electrical crystals across the interface is also used, sometimes utilising beam reflection or refraction across the interface.

SMOKE DENSITY DETECTOR

For fire warning and exhaust gas indication a photo-cell in conjunction with an amplifier and alarm or indicator is used. Three types are in use, those which operate by light scatter, by light obscuration and a combination.

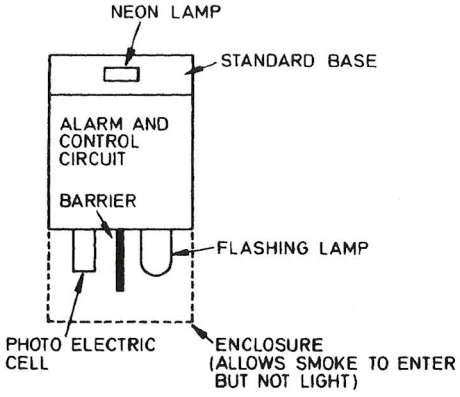

Fig. 5.6 SMOKE DENSITY DETECTOR

A light scatter photo-cell separated from a semi-conductor intermittently flashing light source is shown in Fig. 5.6. The housing enclosure allows smoke but not light inside. With smoke present in the container light is scattered around the barrier on to the photo-cell and an alarm is triggered. The light obscuration type is used in oil mist detection for IC engine crankcases and the combination type is used for detection in CO_2 flooding systems.

OIL MIST DETECTOR

The photo-cells of Fig. 5.7 are normally in a state of electric balance, *i.e.* measure and reference tube mist content in equilibrium. Out of balance current due to rise of crankcase mist density can be arranged to indicate on a galvanometer which can be connected to continuous chart recording and auto visual or audible alarms. The suction fan draws a large volume of slow moving oil-air vapour mixture in turn from various crankcase sel-

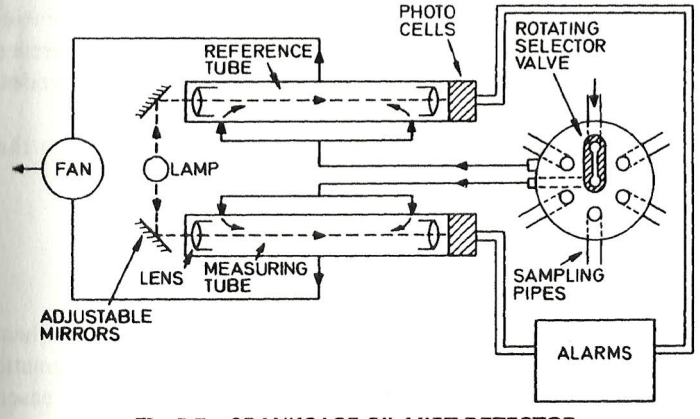

Fig. 5.7 CRANKCASE OIL MIST DETECTOR

ection points. Oil mist near the lower critical density region has a very high optical density. Alarm is normally arranged to operate at 2·5% of the lower critical point, *i.e.* assuming 50 mg/l as lower explosive limit then warning at 1·25 mg/l.

pH SENSOR

The pH value of a solution is the logarithm of the reciprocal of the hydrogen ion concentration in the solution. Its value ranges between 0 to 14, neutrality being 7 anything from 7 to 14 is alkaline and from 0 to 7 is acidic. pH measurement and control is extremely important, being primarily used for feed water analysis.

The method of pH measurement is by means of a conductivity cell consisting of two electrodes and a temperature sensor, pH value varies with temperature hence it is important that this be controlled by means of a sensor/compensator. One of the electrodes is a reference electrode which has a fixed potential irrespective of the variation of hydrogen ion concentration of the solution. The other electrode produces a potential dependent mainly upon the difference in hydrogen ion concentrations between the buffer solution and the solution whose pH has to be measured (across the membrane). In this way the potential difference between the glass measuring electrode and reference electrode is a measure of the pH value of the solution. Electrodes, with sensor/compensator, are inserted in the fluid flow path.

Figure 5.8 shows the two types of electrode used in the conductivity cell.

HEAT (FIRE) DETECTOR

Detector heads are generally one of three types. At a set heat (temperature) condition the increasing pressure on the pneumatic diaphragm bulb type closes electrical alarm contacts, increased differential temperature on the bi-metallic type activates alarms, increased heat fractures a quartzoid bulb (containing a highly expansive fluid) releasing water sprinkler supply and pressure alarm. A typical fire detection-alarm circuit is shown in Fig. 5.9.

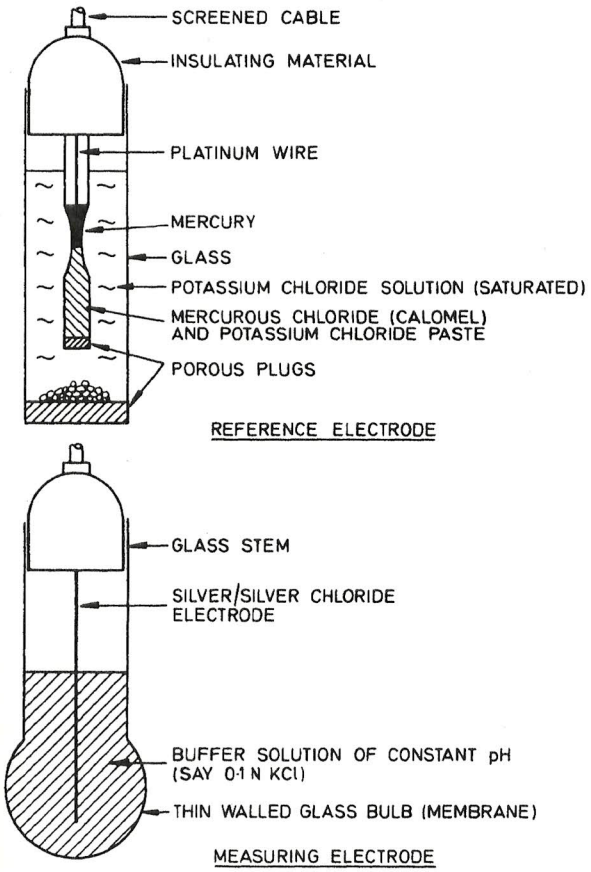

Fig. 5.8 pH SENSOR

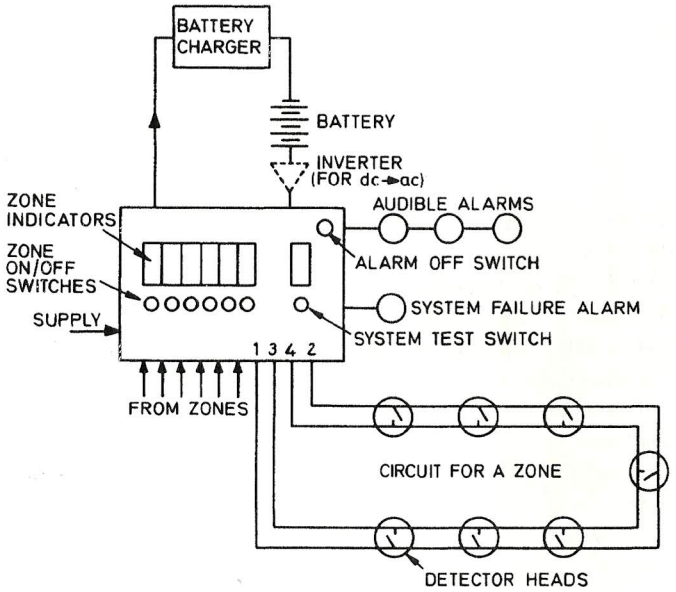

Fig. 5.9. FIRE DETECTION – ALARM CIRCUIT

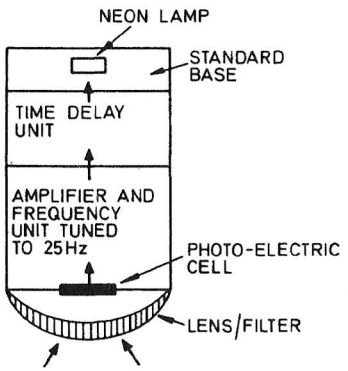

Fig. 5.10 FLAME DETECTOR

FLAME DETECTOR

Fig. 5.10 illustrates the infra red type of device. Flame has a characteristic flicker frequency of about 25 Hz and use is made of this fact to trigger an alarm. Flickering radiation from flames reaches the detector lens/filter unit, which only allows infra-red rays to pass and be focused upon the cell. The signal from the cell goes into the selective amplifier, which is tuned to 25 Hz, then into a time delay unit (to minimise incidence of false alarms, fire has to be present for a pre-determined period), trigger and alarm circuits.

GAS EXPLOSION-DETECTOR METER

The instrument illustrated in Fig. 5.11 is first charged with fresh air from the atmosphere using the rubber aspirator bulb *(A)*. On-off switch *(S_2)* is closed together with check switch *(S_1)* and the compensatory filament *(C)* and detector filament *(D)* allowed to reach steady state working temperature. The zero adjustment rheostat *(F)*, can now be adjusted so that galvanometer *(G)* reads zero. Voltage is adjustable from battery *(B)* by the rheostat *(E)*. Switch S_2 is now opened.

The instrument is now charged from the suspect gas space and while operating the bulb, the switch S_2 is again closed. If a flammable or explosive gas is present it will cause the detector filament to increase in temperature. This disturbs the bridge balance and a current flows. Galvanometer *G* can be calibrated so that the scale is marked to read "% of Lower Limit of Explosive Concentration of Gas".

An alternative design has two ionising chambers, one reference (air) and the other sample, each containing a radioactive ionising source. Combustion particles when ionised are more bulky and less mobile than normal gas molecules so they are readily neutralised. This results in higher resistance and voltage change at the sample chamber — which activates alarms.

GAS ANALYSIS

For detailed gas analysis an Orsat apparatus is used. However a number of measurements require to be continuously recorded. Two representative examples can be considered namely an oxygen analyser and a carbon dioxide analyser.

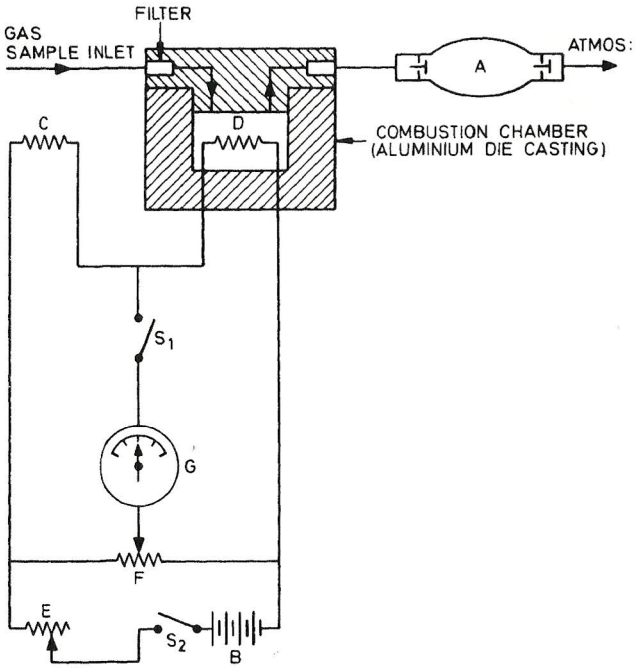

Fig. 5.11 GAS EXPLOSION – DETECTOR METER

Oxygen analyser

Gases can generally be classified as either diamagnetic or paramagnetic, the former seek the weakest part of a magnetic field and the latter the strongest. Most of the common gases are diamagnetic but oxygen is paramagnetic and use is made of this in the oxygen analyser shown in Fig. 5.12.

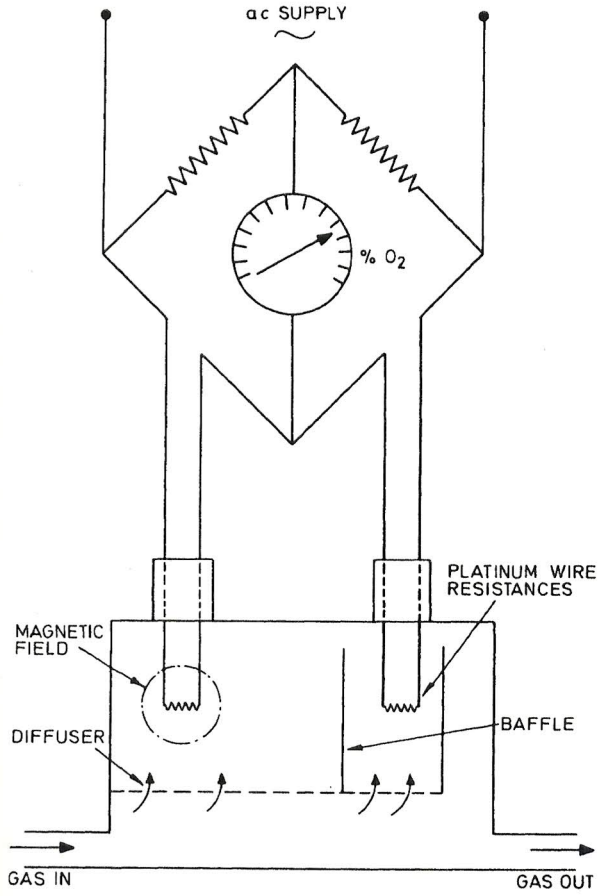

Fig. 5.12 OXYGEN ANALYSER

Two platinum wire resistances are heated by current from an ac bridge and the gas to be measured enters the resistance chamber via a diffuser. One of the resistance wires is placed in a magnetic field hence oxygen is drawn towards this resistance, thus convection currents are set up around this resistance which is then cooled relatively to the other resistance. The bridge is then unbalanced, the amount of unbalance is a measure of the oxygen content and this is displayed on the galvanometer.

CO_2 analyser

Referring to Fig. 5.13.

Approximate thermal conductivities are in proportion:

$$CO_2 = 1,\ H_2O = 1,\ CO = 4,\ O_2 = 2,\ N_2 = 2$$

The sample enters via a filter and drier, water vapour must be removed as it has the same conductivity as CO_2. The wire cell resistance is proportional to heat dissipation, proportional to thermal conductivity of gas in the cell, proportional therefore to CO_2 content. Air is used in the reference cell. Thus the only difference between gas sample and air, from the thermal conductivity viewpoint is CO_2 (as H_2O removed and O_2 and N_2 same value). This assumes no CO or H_2, if these are present (normally only very small proportions) they will be registered as CO_2 unless the sample is first passed over a burner and these two gases burned off before the reading.

Thus the Wheatstone bridge electrical unbalance is dependent on CO_2 content and the unbalance electrical current is measured by the potentiometer.

Chemical absorption and mechanical types are also used.

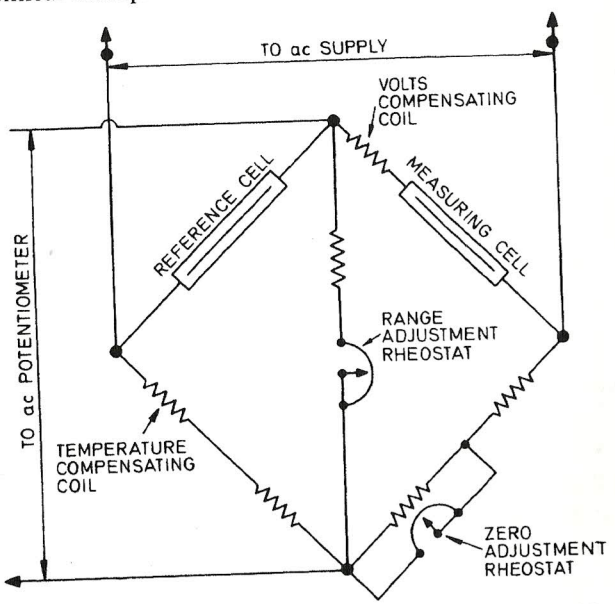

Fig. 5.13 THERMAL CONDUCTIVITY TYPE CO_2 RECORDER

RELATIVE HUMIDITY

A hair element will react to changes of humidity and provide a linear movement, with negligible force, which can be converted to electrical or pneumatic signal and amplified as required.

WATER ANALYSIS

The measurement of pH has been considered previously. Two other measurements are commonly required, *i.e.* electrical conductivity meter for dissolved solid assessment and dissolved oxygen meter.

"Dionic" water purity meter

Specific conductivity mho/cm³ is the conductance of a column of mercury 1 cm² cross sectional area and 1 cm long. This is a large unit and micromho/cm³ (reciprocal megohm) is used and when corrected to 20°C is called a dionic unit. Conductivity of pure distilled water is about 0·5, fresh water about 500 dionic units. The sensor, shown in Fig. 5.14, measures conductivity of

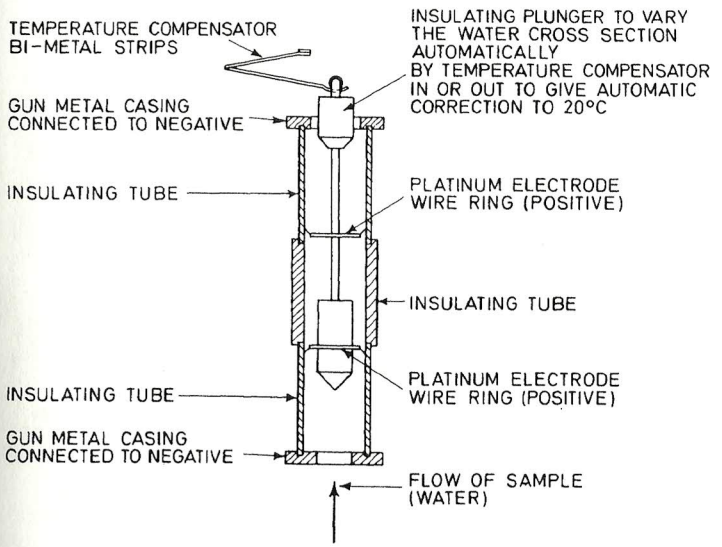

Fig. 5.14 DIONIC WATER PURITY METER

two water columns in parallel, *i.e.* between positive platinum rings and negative gunmetal collars. The insulating plunger, operated by bi-metallic strip, varies cross sectional area for automatic correction to 20°C. The measurement is by conventional ohmeter. The device should be used with de-gassifying units to avoid errors due to occlusion of carbon dioxide.

Dissolved oxygen meter

The unit is shown in Fig. 5.15. The sample water flows via a chamber which surrounds the katharometer (Wheatstone bridge circuit) and receives pure hydrogen. Some hydrogen is taken into solution and this releases some dissolved oxygen (in air). This mixture passes to atmosphere across one side of the bridge whilst the other side is in pure hydrogen. The cooling effect is different on the two sides of the katharometer, depending on air (oxygen) present, and resultant unbalance current operates indicator or recorder calibrated directly in ppm oxygen. For very low oxygen content it is often necessary to utilise an electro-chemical cell in place of this meter.

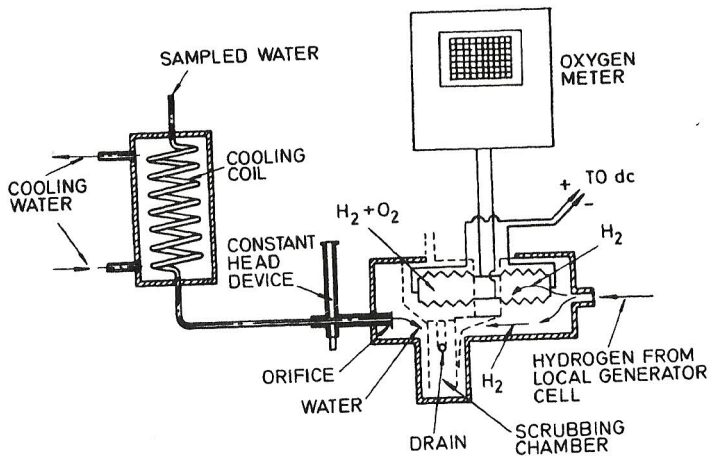

Fig. 5.15 DISSOLVED OXYGEN METER

INSTRUMENT CALIBRATION. TESTING AND ADJUSTMENT

Generally a specialist subject. Pneumatic instruments would be tested by master gauge, standard manometer or deadweight tester. Electrical instruments are tested by standard resistors, potentiometers, capacitors, etc.

Using a Bourdon pressure gauge as example:

1. Zero (error) adjustment changes base point without changing the slope or shape of the calibration curve. It is usually achieved by rotating the indicator pointer relative to the movement, linkage and element
2. Multiplication (magnification) adjustment alters the slope without changing base point or shape. This is effected by altering the drive linkage length ratios between primary element and indicator pointer.
3. Angularity adjustment changes the curve shape without altering base point and alters scale calibration at the ends. This error is minimised by ensuring that link arms are perpendicular with the pointer at mid scale.

Fig. 5.16 shows calibration curves and adjustment for the Bourdon link type of instrument mechanism.

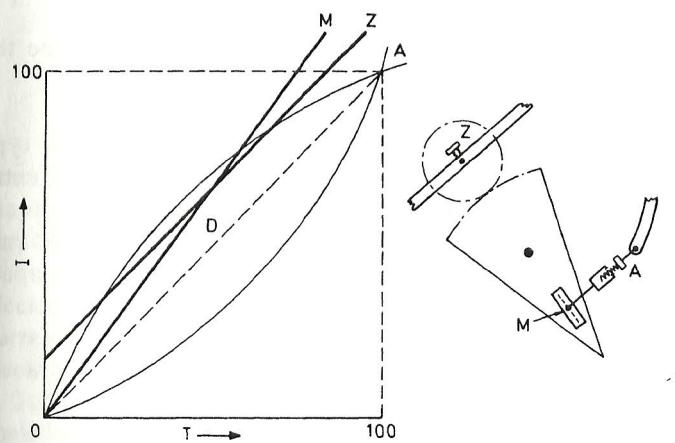

Fig. 5.16 INSTRUMENT CALIBRATION AND ADJUSTMENT

Instrument readings (I), true values (T), desired result (D). Zero error and adjustment (Z), multiplication error and adjustment (M), angularity error and adjustment (A) — error curves or lines for actual values.

Over the design range pointer movement bears a linear relationship to pressure, and the scale is calibrated accordingly.

Hysteresis — a vibration phenomena. Best eliminated by correctly meshed gearing and fitted pivots to reduce backlash, etc.

TEST EXAMPLES 5

1. Explain the principle of operation of a carbon dioxide recorder for monitoring the uptake gases.
State what normal maintenance it requires.
State what action is taken if the carbon dioxide content is unacceptably low.
If this action does not alter the carbon dioxide reading, explain how the accuracy of the recorder is checked and adjusted.

2. Explain the term "photo-electric effect" and describe equipment suitable for crankcase monitoring and fire detection in which this phenomenon is utilised.

3. Describe, with the aid of sketches, a torsionmeter.
Explain the principle of operation and briefly describe the construction of the shaft and indicator units.

4. Discuss the errors liable to be exhibited by link type instruments. Describe how such an instrument could be calibrated and adjusted to reduce these errors to a minimum.

CHAPTER 6

TELEMETERING

The objective of this chapter (and Chapter 7) is to link instrumentation in preceding chapters, primarily concerned with measuring devices, with control elements as described in subsequent chapters. Some repetition may inevitably result in the presentation.

There are a wide range of variables to be measured. Detecting and measuring elements are mainly electrical but a significant number are displacement operated mechanical types. Chemical and electronic devices are also used.

Transducers can generally be simplified into three basic reversible types namely:

mechanical displacement↔pneumatic

mechanical displacement↔electrical

pneumatic↔electrical

Pneumatic principles are invariably flapper nozzle; orifice, diaphragm. Electrical principles include resistance change, variable inductance, variable capacitance, current or voltage, with frequency and phase used to a limited extent. Conversion of electrical signal is also used, *i.e.* resistance-current, voltage-current, etc. and such modern transducers often incorporate electronic oscillators and amplifiers.

Telemetering may be defined as signal transmission over a considerable distance. The device at the measure point, usually a transducer, is then often called a transmitter with the receiver located at the recording or control centre.